

Balanced deficiency rotors in shortest-support Hall systems of triangle-free maximum cuts

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Abstract

Fix a maximum cut of a triangle-free graph, write B for its crossing edges and M for its monochromatic edges, and associate to each $xy \in M$ with $d_B(x, y) = 4$ the crossing edges on its shortest geodesics. The companion note [6] showed that the resulting supports can violate Hall's condition by an unbounded factor and classified the minimal violating footprints with at most ten atoms; the unique nine-atom footprint found there is the smallest object studied here. This paper treats the dynamic side of the same system. We prove an abstract circuit identity for inclusion-minimal Hall violations (support union one edge smaller than the atom set, with a perfect system of distinct representatives after deleting any atom), a star lemma at cut-tight vertices whose arithmetic core is kernel-checked in Lean 4, and an eight-vertex example, the neutral square rotor, on which geodesic-collision repair provably never terminates even though Hall expansion holds with room to spare. We then close the two smallest rotor windows of the associated transport strategy: nine-atom support circuits carry no balanced rotor (two independent proofs, with kernel-checked finite cores), and at sixteen atoms no covered two-owner window exists (an exhaustive, dually verified census of 153,978 support graphs). At twenty-five atoms the results are partial: support orders 15 and 16 are excluded by machine certificates, the admissible order window is 15–21, and two eighteen-vertex circuits that pass every local test are excluded only by explicit certificates; orders 17 and 19–21 remain open. All computer-assisted claims carry named verifiers and SHA-256 hashes. These results constrain one local proof strategy; they yield no upper bound on $\text{bip}(G)$ and do not resolve the Erdős $n^2/25$ conjecture.

1 Introduction

For a finite graph G , let $\text{bip}(G) = |E(G)| - \text{MaxCut}(G)$ be its edge-deletion distance to bipartiteness. Erdős, Faudree, Pach and Spencer [3] asked whether every triangle-free N -vertex graph satisfies $\text{bip}(G) \leq N^2/25$; the balanced blow-up of C_5 shows that the constant would be sharp. See [4, 1] for the strongest general bounds, and [5] for an exact certificate of the conjectured value at $N = 5n$, $1 \leq n \leq 40$. Nothing in the present paper improves any of these bounds.

The companion note [6] introduced the shortest-geodesic support system of a fixed cut: each monochromatic (*bad*) edge xy with $d_B(x, y) = 4$ is charged to the crossing edges on its shortest x – y paths, and one asks when these supports admit a system of distinct representatives, i.e. satisfy Hall's condition. The note constructed triangle-free graphs with unique maximum cuts on which the Hall condition fails by an unbounded factor, and classified the minimal violating footprints exactly: none with at most eight atoms, a unique nine-atom footprint (a double star whose atom set is a $K_{3,3}$), and three ten-atom classes.

This paper is the dynamic companion to that static classification. If supports fail Hall expansion, a natural repair strategy selects one geodesic per bad edge and locally re-routes colliding selections by *detour moves*; the terminal objects such a transport strategy must exclude

are configurations in which a locally deficient pattern survives every move by merely rotating — *balanced rotors*. Our results are of two kinds.

Structures. Section 2 isolates the part of the companion’s footprint lemma that depends on inclusion-minimality alone: a support circuit on m atoms has exactly $m - 1$ support edges, every edge lies in at least two supports, the atom–edge incidence graph is connected, and deleting any single atom leaves a perfect system of distinct representatives; in classical terms, support circuits are circuits of a transversal matroid. The nine-atom footprint of [6] is exactly the $t = 3$ instance of these circuits, at t^2 atoms on $t^2 - 1$ support edges. Section 3 proves an independent local fact about maximum cuts of triangle-free graphs, the star lemma at cut-tight vertices, whose arithmetic core and pair manipulations are kernel-checked in Lean 4. Section 4 constructs an eight-vertex *neutral square rotor*: a triangle-free graph with a maximum cut on which every bad edge has two geodesics and Hall expansion holds strictly, yet no selection is collision-free and the repair walk never terminates — its four states form a single orbit of a cut-preserving automorphism.

Closures. Section 5 proves, in two independent ways, that nine-atom support circuits carry no balanced rotor: a shape-independent support-edge count, and a coverage-emptiness argument on the unique nine-atom footprint of [6]; both have kernel-checked Lean 4 cores. Section 6 isolates the first surviving window beyond nine atoms — sixteen atoms with two rotating owners of width four, left a margin of exactly one support edge by the by-hand crossover bounds — and closes it by an exhaustive census of all 153,978 connected bipartite graphs with fifteen edges, verified along two acceptance paths. Section 7 collects the partial state of the twenty-five-atom window: an order window $15 \leq |V(F)| \leq 21$ proved by hand, machine exclusion of orders 15 and 16, two certified eighteen-vertex near-misses, and an explicit open question. The closures at $t = 3, 4$ are complete; $t = 5$ is not.

Every computer-assisted claim names its verifier and the SHA-256 hash of the artifact or source it rests on, with single-solver certificates flagged as such; Section 8 aggregates the verification record, and the ancillary files contain the verifier scripts and the cited Lean 4 sources. We stress the scope honestly: the theorems below constrain one local proof strategy for the $N^2/25$ conjecture — they eliminate specific repair mechanisms, at specific window sizes — and they produce no new upper bound on $\text{bip}(G)$ and no progress on the conjecture itself.

2 Support circuits

Fix a cut $V(G) = V_0 \dot{\cup} V_1$ of a triangle-free graph G , write B for the crossing edges and $M = E(G) \setminus B$ for the monochromatic edges, and for $xy \in M$ with $d_B(x, y) = 4$ let $\text{supp}_B(xy) \subseteq B$ be the set of edges lying on shortest x - y paths in B , as in the companion paper [6]. A failure of the Hall condition $|\text{supp}_B(\mathcal{S})| \geq |\mathcal{S}|$ localizes to an inclusion-minimal violating family, and the Minimal footprint lemma of [6] lists the graph-level constraints such a family must satisfy. In this section we isolate the part of that lemma that depends on inclusion-minimality alone. It is a statement about abstract finite set systems; no graph structure enters. Besides recovering the counting and multiplicity clauses of the footprint lemma, the abstract form yields two further constraints used repeatedly in later sections: connectivity of the atom–edge incidence graph, and a family of perfect systems of distinct representatives obtained by deleting any single atom. All proofs are elementary and complete.

2.1 The abstract identity

Let A be a finite index set, whose elements we call *atoms*, and let $\mathcal{F} = (F_a)_{a \in A}$ be a family of finite sets. For $S \subseteq A$ write

$$F(S) = \bigcup_{a \in S} F_a, \quad F^* = F(A).$$

In the intended application the atoms are bad edges $xy \in M$ with $d_B(x, y) = 4$ and $F_a = \text{supp}_B(a)$, but nothing below uses this.

Definition 2.1. The family $\mathcal{F}$ is *deficient* if $|F^*| < |A|$. It is a *support circuit* if it is deficient and every proper subfamily satisfies the Hall condition:

$$|F(S)| \geq |S| \quad \text{for every } S \subsetneq A.$$

Support circuits are exactly the inclusion-minimal deficient families: a deficient proper subfamily is precisely a violation of the displayed condition.

Theorem 2.2 (Circuit identity). *Let $\mathcal{F} = (F_a)_{a \in A}$ be a support circuit with $m = |A| \geq 2$. Then:*

- (i) $|F^*| = m - 1$;
- (ii) $F(A \setminus \{a\}) = F^*$ for every $a \in A$;
- (iii) every $f \in F^*$ lies in at least two of the sets F_a ;
- (iv) the bipartite incidence graph on $A \dot{\cup} F^*$, with a adjacent to f when $f \in F_a$, is connected;
- (v) for every $a \in A$ there is a bijection $\varphi_a: A \setminus \{a\} \rightarrow F^*$ with $\varphi_a(b) \in F_b$ for all $b \neq a$.

Proof. (i), (ii). Fix $a \in A$. The proper subfamily $A \setminus \{a\}$ satisfies the Hall condition, so

$$m - 1 \leq |F(A \setminus \{a\})| \leq |F^*| \leq m - 1,$$

the last inequality by deficiency and integrality. Equality holds throughout; in particular $|F^*| = m - 1$, and $F(A \setminus \{a\})$ is a subset of F^* of full cardinality, hence equal to it.

(iii). Let $f \in F^*$, say $f \in F_a$. By (ii), $f \in F(A \setminus \{a\})$, so $f \in F_b$ for some $b \neq a$.

(iv). Since $m \geq 2$, every singleton $\{a\}$ is a proper subfamily, so $|F_a| \geq 1$; and every $f \in F^*$ lies in some F_a . Hence every connected component of the incidence graph contains at least one atom. Suppose there are two or more components. Splitting the components into two nonempty groups partitions A into nonempty sets A_1, A_2 with $F(A_1) \cap F(A_2) = \emptyset$: an element in both unions would be adjacent to atoms in distinct components. Both A_i are proper subfamilies, so

$$|F^*| = |F(A_1)| + |F(A_2)| \geq |A_1| + |A_2| = m,$$

contradicting (i).

(v). Every $S \subseteq A \setminus \{a\}$ is a proper subfamily of A , so $|F(S)| \geq |S|$. By Hall's theorem [7] the family $(F_b)_{b \in A \setminus \{a\}}$ has a system of distinct representatives: an injection φ_a with $\varphi_a(b) \in F_b$. Its image is a subset of F^* of size $m - 1 = |F^*|$, so φ_a is a bijection onto F^* . $\square$

Remark 2.3. The identity in (i) is intrinsic to the circuit: if a support circuit A sits inside a larger ambient family $\mathcal{M}$, then $|F^*| = |A| - 1$, not $|\mathcal{M}| - 1$. All counting in later sections is against the circuit size m .

Remark 2.4. The hypothesis $m \geq 2$ only excludes the degenerate circuit $A = \{a\}$, $F_a = \emptyset$, for which (i), (ii) still hold trivially. In the geodesic application $m \geq 5$ (Corollary 2.8).

Clause (v) says more than the existence of a matching: after deleting *any* one atom, the remaining $m - 1$ atoms match onto *all* of F^* , each atom represented inside its own set. This rigidity is the main tool of the window analyses in later sections. It also identifies support circuits in classical terms.

Corollary 2.5. *A family $(F_a)_{a \in A}$ is a support circuit if and only if it has no system of distinct representatives while every proper subfamily has one.*

Proof. By Hall's theorem, a finite family has a system of distinct representatives if and only if all of its subfamilies satisfy the Hall condition.

Let A be a support circuit. Deficiency is a violation of the Hall condition at $S = A$, so A has no system of distinct representatives. Every subfamily of a proper subfamily of A is itself a proper subfamily of A and hence satisfies the Hall condition, so every proper subfamily has a system of distinct representatives.

Conversely, suppose A has no system of distinct representatives while every proper subfamily has one. Some $S \subseteq A$ violates the Hall condition; S cannot be proper, since a proper subfamily has a system of distinct representatives and therefore satisfies $|F(S)| \geq |S|$. Hence A itself is deficient, and every proper subfamily satisfies the Hall condition. $\square$

Remark 2.6. The subfamilies of an ambient family $\mathcal{M}$ admitting systems of distinct representatives are the independent sets of a matroid, the transversal matroid induced by $(F_a)_{a \in \mathcal{M}}$ [2]. Corollary 2.5 says that the support circuits inside $\mathcal{M}$ are exactly the circuits of this matroid, and Theorem 2.2(v) computes the rank of a circuit deleted by one element: the deletion matchings are perfect onto the ground union F^* .

2.2 The graph form

Specializing to shortest-geodesic supports recovers the Minimal footprint lemma of [6] and extends it by the two new clauses. We restate the full result in the paper's setting; the genuinely graph-level clauses (c), (d) have the same short proofs as in [6] and are included for completeness.

Corollary 2.7 (Graph support circuits). *Let G be triangle-free with a fixed cut, let B be connected, and let $\mathcal{A} \subseteq \{xy \in M : d_B(x, y) = 4\}$ be inclusion-minimal with*

$$|\text{supp}_B(\mathcal{A})| < |\mathcal{A}| = m.$$

Write $F = \text{supp}_B(\mathcal{A})$, viewed as the subgraph it spans. Then:

- (a) $|E(F)| = m - 1$, and F is connected and bipartite;
- (b) every edge of F lies in at least two atom supports, and $\text{supp}_B(\mathcal{A} \setminus \{a\}) = E(F)$ for every $a \in \mathcal{A}$;
- (c) every $xy \in \mathcal{A}$ satisfies $d_F(x, y) = 4$, and $\text{supp}_B(xy)$ is the union of all length-four x - y paths in F ;
- (d) the graph $(V(F), \mathcal{A})$ is triangle-free;
- (e) for every $a \in \mathcal{A}$ there is a bijection from $\mathcal{A} \setminus \{a\}$ onto $E(F)$ assigning to each atom an edge of its own support.

Proof. The family $(\text{supp}_B(a))_{a \in \mathcal{A}}$ is a support circuit, and every support is nonempty because $d_B(x, y) = 4$, so $m \geq 2$ and Theorem 2.2 applies with $F^* = E(F)$.

(a) The count is Theorem 2.2(i), and $F \subseteq B$ is bipartite. For connectivity: each support $\text{supp}_B(xy)$ is a connected edge set, being a union of x - y paths that all contain x . If two atoms are adjacent to a common edge f in the incidence graph, their supports share the edge f and hence their union is connected. Walking along paths of the incidence graph, which is connected by Theorem 2.2(iv), connects all supports, and every edge of F lies in some support.

(b) Theorem 2.2(iii) and (ii).

(c) The edges of any shortest x - y path in B lie in $\text{supp}_B(xy) \subseteq E(F)$, so $d_F(x, y) \leq 4$. Since $F \subseteq B$ and x, y lie on the same shore, every x - y walk in F has even length, and $d_F(x, y) = 2$ would give a common neighbour forming a triangle with the edge xy . Hence $d_F(x, y) = 4$. A length-four x - y path in F is a length-four x - y path in B , hence shortest, hence contained in

$\text{supp}_B(xy)$; conversely every shortest x - y path in B lies in $\text{supp}_B(xy) \subseteq E(F)$ and has length $4 = d_F(x, y)$, so it is one of the length-four F -paths.

(d) $\mathcal{A} \subseteq E(G)$ and G is triangle-free; the endpoints of atoms lie in $V(F)$ by (c).

(e) Theorem 2.2(v). □

Corollary 2.8 (First sizes). *In the setting of Corollary 2.7, $m \geq 5$, and the average multiplicity of a support edge satisfies*

$$\frac{1}{|E(F)|} \sum_{a \in \mathcal{A}} |\text{supp}_B(a)| \geq \frac{4m}{m-1} > 4.$$

Proof. Each support contains the four edges of a shortest path, so $4 \leq |\text{supp}_B(a)| \leq |E(F)| = m-1$, giving $m \geq 5$; and $\sum_a |\text{supp}_B(a)| \geq 4m$ with $|E(F)| = m-1$. □

Remark 2.9. Corollary 2.7(a)–(d) recovers the Minimal footprint lemma of [6], where it feeds the exact classification of the footprints with $m \leq 10$; clause (e), the incidence connectivity behind (a), and the deletion-union identity in (b) (which occurs in the proof, but not the statement, of the companion lemma) are the additional structure carried by the circuit form. Sharper counts are derived in later sections for *selected* supports, in which each atom retains a single geodesic; these depend on the owner–profile apparatus. The results of this section constrain the geometry of a minimal Hall violation; they give no upper bound on $\text{bip}(G)$.

3 A star lemma at cut-tight vertices

This section records a self-contained fact about maximum cuts of triangle-free graphs. It uses only triangle-freeness and maximality of the cut, and is independent of the support and Hall systems studied in the rest of the paper.

Throughout the section, G is a finite simple graph with a fixed cut $V(G) = V_0 \dot{\cup} V_1$, B is the set of crossing edges, and $M = E(G) \setminus B$ is the set of monochromatic edges. The cut is a *maximum cut* when $|M|$ is least over all cuts of G , so that $|M| = \text{bip}(G)$.

Definition 3.1 (Switch loss; cut-tight vertices). For $S \subseteq V(G)$, let $\partial_B S$ (respectively $\partial_M S$) be the set of edges of B (respectively M) with exactly one endpoint in S , and define the *switch loss*

$$\sigma(S) = |\partial_B S| - |\partial_M S|.$$

For a vertex v , write $\sigma(v) = \sigma(\{v\}) = d_B(v) - d_M(v)$, where $d_B(v)$ and $d_M(v)$ count the crossing and monochromatic edges at v . We call v *cut-tight* if $\sigma(v) \leq 1$, and we write $N_B(v)$ for the set of neighbours joined to v by an edge of B , so $|N_B(v)| = d_B(v)$.

Lemma 3.2 (Switch identity). *Exchanging the shores of the vertices of a set $S \subseteq V(G)$ produces a cut with exactly $|M| + \sigma(S)$ monochromatic edges. Consequently, at a maximum cut,*

$$\sigma(S) \geq 0 \quad \text{for every } S \subseteq V(G),$$

and in particular $\sigma(v) \geq 0$ for every vertex v .

Proof. An edge changes its crossing status under the switch if and only if exactly one of its endpoints lies in S ; all other edges keep their status. The new monochromatic set is therefore $(M \setminus \partial_M S) \cup \partial_B S$, of size $|M| - |\partial_M S| + |\partial_B S| = |M| + \sigma(S)$. A maximum cut minimizes the monochromatic count, so $\sigma(S) \geq 0$. □

Lemma 3.3 (Additivity on a nonedge). *If $x \neq y$ and $xy \notin E(G)$, then for any cut*

$$\sigma(\{x, y\}) = \sigma(x) + \sigma(y).$$

Proof. Since $xy \notin E(G)$, no edge is incident to both x and y . Hence an edge has exactly one endpoint in $\{x, y\}$ if and only if it is incident to exactly one of x, y , and $\partial_B\{x, y\} = \partial_B\{x\} \dot{\cup} \partial_B\{y\}$, $\partial_M\{x, y\} = \partial_M\{x\} \dot{\cup} \partial_M\{y\}$. Subtract. $\square$

Lemma 3.4 (Star inequality). *Let G be triangle-free, fix any cut, and let v be a vertex with $k = d_B(v)$. Then*

$$\sigma(\{v\} \cup N_B(v)) = \sigma(v) + \sum_{a \in N_B(v)} \sigma(a) - 2k,$$

and hence, at a maximum cut,

$$\sum_{a \in N_B(v)} \sigma(a) \geq 2d_B(v) - \sigma(v). \quad (1)$$

Proof. Set $S = \{v\} \cup N_B(v)$. If two vertices $a, a' \in N_B(v)$ were adjacent, then va, va', aa' would form a triangle; so $N_B(v)$ is independent. An edge of G with both endpoints in S therefore joins v to some $a \in N_B(v)$, and since G is simple this edge is the crossing edge va itself. Thus the edges of G inside S are exactly the k crossing star edges, and no monochromatic edge has both endpoints in S . Summing degrees over S counts each internal edge twice and each boundary edge once:

$$|\partial_B S| = \sum_{u \in S} d_B(u) - 2k, \quad |\partial_M S| = \sum_{u \in S} d_M(u).$$

Subtracting gives the identity, and Lemma 3.2 gives (1) at a maximum cut. $\square$

At a cut-tight vertex, (1) reads $\sum_{a \in N_B(v)} \sigma(a) \geq 2k - 1$: the k crossing neighbours carry average switch loss at least $2 - 1/k$. Two immediate consequences:

Corollary 3.5. *At a maximum cut, every cut-tight vertex satisfies $d_M(v) \geq d_B(v) - 1$. In particular, a cut-tight vertex with $d_B(v) \geq 2$ is an endpoint of a monochromatic edge.*

Proof. $d_B(v) - d_M(v) = \sigma(v) \leq 1$. $\square$

Corollary 3.6. *At a maximum cut of a triangle-free graph, every cut-tight vertex v with $d_B(v) \geq 2$ has a crossing neighbour $a \in N_B(v)$ with $\sigma(a) \geq 2$.*

Proof. Write $k = d_B(v) \geq 2$. If every $a \in N_B(v)$ had $\sigma(a) \leq 1$, then $\sum_{a \in N_B(v)} \sigma(a) \leq k \leq 2k - 2 < 2k - 1$, contradicting (1) with $\sigma(v) \leq 1$. $\square$

A single high-loss neighbour already pairs with the whole star:

Proposition 3.7 (Pair threshold). *Let G be triangle-free with a maximum cut, and let x, y be distinct crossing neighbours of a common vertex. If $\sigma(x) \geq 2$, then $xy \notin E(G)$ and*

$$\sigma(\{x, y\}) = \sigma(x) + \sigma(y) \geq 2.$$

Proof. Nonadjacency is triangle-freeness, as in the proof of Lemma 3.4. By Lemma 3.3, $\sigma(\{x, y\}) = \sigma(x) + \sigma(y)$, and $\sigma(y) \geq 0$ by Lemma 3.2. $\square$

The main statement refines Corollary 3.6: the combined loss of two neighbours can be forced relative to any prescribed neighbour. The arithmetic is a pigeonhole, which we isolate in the form in which it has been formally verified.

Lemma 3.8 (Star pigeonhole). *Let A be a finite set with $|A| \geq 2$, let $\ell: A \rightarrow \mathbb{Z}_{\geq 0}$, and let $s \in \{0, 1\}$ satisfy*

$$2|A| \leq s + \sum_{a \in A} \ell(a).$$

Then for every $x \in A$ there exists $y \in A \setminus \{x\}$ with $\ell(x) + \ell(y) \geq 2$.

Proof. Suppose $\ell(x) + \ell(y) \leq 1$ for every $y \in A \setminus \{x\}$. Since $|A| \geq 2$, some $y_0 \in A \setminus \{x\}$ exists; from $\ell(x) + \ell(y_0) \leq 1$ and $\ell \geq 0$ we get $\ell(x) \leq 1$, and likewise $\ell(y) \leq 1$ for every $y \in A \setminus \{x\}$. Hence $\sum_{a \in A} \ell(a) \leq |A|$ and $2|A| \leq s + |A| \leq 1 + |A|$, so $|A| \leq 1$, a contradiction. $\square$

Theorem 3.9 (Star lemma at cut-tight vertices). *Let G be triangle-free, fix a maximum cut, and let v be a cut-tight vertex with $d_B(v) \geq 2$. Then for every $x \in N_B(v)$ there exists $y \in N_B(v) \setminus \{x\}$ with $xy \notin E(G)$ and*

$$\sigma(\{x, y\}) = \sigma(x) + \sigma(y) \geq 2.$$

Proof. Apply Lemma 3.8 with $A = N_B(v)$, $\ell = \sigma$ restricted to A (nonnegative by Lemma 3.2), and $s = \sigma(v) \in \{0, 1\}$; hypothesis (1) is the star inequality. This yields $y \in N_B(v) \setminus \{x\}$ with $\sigma(x) + \sigma(y) \geq 2$. Since x and y are distinct crossing neighbours of v and G is triangle-free, $xy \notin E(G)$, and Lemma 3.3 gives $\sigma(\{x, y\}) = \sigma(x) + \sigma(y)$. $\square$

Remark 3.10. Theorem 3.9 is a necessary condition satisfied by every maximum cut of every triangle-free graph. By itself it yields no upper bound on $\text{bip}(G)$ and no progress on the Erdős $N^2/25$ conjecture; in this paper it serves only as a local production threshold: inside the crossing star of any cut-tight vertex, every neighbour can be paired to a nonadjacent partner whose joint switch loss meets the threshold two.

Remark 3.11 (Formalization). Four ingredients of this section are verified in Lean 4, in the development `Erdos23Delta0` accompanying this project. There, graphs are literal edge lists with executable adjacency, cuts are side assignments, and a maximum cut is a side assignment minimizing the monochromatic count over all side assignments.

- (i) Singleton nonnegativity (the vertex case of Lemma 3.2): `singleton_sigma_nonneg_of_isMaxCut` in the module `Erdos23Delta0.Gamma.SingletonPairSigma`; the general-set inequality $\sigma(S) \geq 0$ is `sigmaNonneg_of_badCount_min` in `Erdos23Delta0.CertGraph`.
- (ii) Additivity on a nonedge (Lemma 3.3): `sigma_pair_eq_add_singletons_of_nonadjacent` in `Erdos23Delta0.Gamma.SingletonPairSigma`, with the crossing and monochromatic boundary counts added separately (`dB_pair_eq_add_singletons_of_nonadjacent`, `dM_pair_eq_add_singl`).
- (iii) The pair threshold (Proposition 3.7): `nonadjacent_of_common_blue`, `two_le_sigma_pair_of_two_le_left`, and their combination `common_blue_pair_two_le_of_left_loss`, all in `Erdos23Delta0.Gamma.Singleton`.
- (iv) The pigeonhole (Lemma 3.8): `exists_other_with_two_le_loss_sum` in `Erdos23Delta0.Gamma.CutTightS` stated over an abstract finite neighbour set with $\mathbb{N}$ -valued losses and the star inequality as a hypothesis (the formal statement does not even require the prescribed element to lie in the set).

The geometric step, the star identity of Lemma 3.4, is proved above but is not among the formalized statements; consequently Theorem 3.9 as a single graph-level statement is proved here on paper, with its arithmetic core and all pair manipulations machine-checked.

4 An eight-vertex neutral square rotor

As before, we fix a cut $V(G) = V_0 \dot{\cup} V_1$ of a triangle-free graph G , write B for the crossing edges and M for the monochromatic (*bad*) edges, and for $e = xy \in M$ with $d_B(x, y) = 4$ let $\text{supp}_B(e)$ be the set of crossing edges lying on shortest x - y paths in B .

The obstructions considered so far are static: sets of bad edges whose support union is too small. This section records a complementary, dynamic phenomenon on eight vertices. In the example, support Hall expansion holds with room to spare and every bad edge even has two geodesics; yet no choice of one geodesic per bad edge yields vertex-disjoint paths, and the natural

local repair walk, which replaces a colliding geodesic by an alternative geodesic of the same bad edge, never terminates: the states form a four-cycle and every state carries a collision. Every statement in this section is an elementary finite check, proved in full below; all of them have in addition been verified exhaustively by machine (Remark 4.8).

Definition 4.1 (The rotor graph). Let R be the graph on the eight vertices a, b, p, q, x, y, m, v with the ten edges

$$ax, yb, pm, vq, \quad xm, my, yv, vx, \quad ab, pq,$$

equipped with the cut $V_0 = \{x, y, p, q\}$, $V_1 = \{m, v, a, b\}$. The first eight edges cross the cut, so $B = \{ax, yb, pm, vq, xm, my, yv, vx\}$ and $M = \{ab, pq\}$. The four edges xm, my, yv, vx form a four-cycle of B , the *square*; the pendant edges attach a, b to the opposite corners x, y and p, q to the opposite corners m, v .

Proposition 4.2. R is triangle-free, B is connected, and the cut of Definition 4.1 is a maximum cut:

$$\text{MaxCut}(R) = 8, \quad \text{bip}(R) = 2,$$

with bad edges exactly ab and pq .

Proof. B is bipartite between V_0 and V_1 , so a triangle of R must use ab or pq ; these two edges are vertex-disjoint, so it uses exactly one. But $N(a) \cap N(b) = \{x, b\} \cap \{y, a\} = \emptyset$ and $N(p) \cap N(q) = \{m, q\} \cap \{v, p\} = \emptyset$. Connectivity of B is clear: every vertex reaches the square by at most one pendant edge.

For the maximum cut, consider the four pentagons

$$C_1 = axmyba, \quad C_2 = axvyba, \quad C_3 = pmxvqp, \quad C_4 = pmyvqp.$$

Every edge of R lies in exactly two of them: ab, ax, yb in C_1, C_2 ; pq, pm, vq in C_3, C_4 ; and the square edges $xm \in C_1, C_3$; $my \in C_1, C_4$; $yv \in C_2, C_4$; $vx \in C_2, C_3$. A cut crosses at most four edges of a five-cycle, so for every cut, counting crossing edges once per containing pentagon,

$$2 \cdot |\{\text{crossing edges}\}| = \sum_{i=1}^4 |\{\text{crossing edges on } C_i\}| \leq 4 \cdot 4 = 16.$$

Hence $\text{MaxCut}(R) \leq 8$. The displayed cut crosses all eight edges of B , so $\text{MaxCut}(R) = 8$ and $\text{bip}(R) = 2$. $\square$

Lemma 4.3 (Geodesic families). *In the fixed cut, both bad edges have B -distance four. Each has exactly two B -geodesics,*

$$A_m = axmyb, \quad A_v = axvyb \quad \text{for } ab,$$

$$B_x = pmxvq, \quad B_y = pmyvq \quad \text{for } pq.$$

Consequently $|\text{supp}_B(ab)| = |\text{supp}_B(pq)| = 6$ and $\text{supp}_B(M) = B$, so the support Hall condition holds strictly for every nonempty subset of M .

Proof. The vertices a, b, p, q have unique B -neighbours x, y, m, v respectively. There is no a - b path in B of length at most three: length two needs a common neighbour, and $N_B(a) \cap N_B(b) = \{x\} \cap \{y\} = \emptyset$; length three needs $w \in N_B(x) \cap N_B(b) = \{a, m, v\} \cap \{y\} = \emptyset$. A length-four path must run $axwyb$ with $w \in N_B(x) \cap N_B(y) \setminus \{a, b\} = \{m, v\}$, giving exactly A_m and A_v . Symmetrically, an p - q geodesic runs $pmwvq$ with $w \in N_B(m) \cap N_B(v) \setminus \{p, q\} = \{x, y\}$, giving B_x and B_y . The two geodesics of ab share the edges ax, yb , so their union has six edges; likewise for pq with pm, vq . The two six-edge supports together cover all eight edges of B . $\square$

Definition 4.4 (Selections, detour moves, owners). Let (V_0, V_1) be a cut in which every bad edge has B -distance four. A *selection* ω assigns to each bad edge one of its B -geodesics; its *selected support* $S_\omega \subseteq B$ is the union of the chosen edge sets. A *detour move* at a bad edge e replaces $\omega(e)$ by a different geodesic of e , leaving all other choices unchanged. A vertex lying on the selected geodesics of two distinct bad edges is an *owner* of ω ; the selection is *disjoint* if it has no owners, that is, if the selected geodesics are pairwise vertex-disjoint.

A disjoint selection upgrades the edge-counting resource measured by supp_B to a packing of vertex-disjoint odd-cycle witnesses, one per bad edge. A natural way to look for one is local repair: while two selected geodesics share a vertex, apply a detour move to one of them. The rotor graph shows that this repair loop need not terminate, even when every bad edge offers a spare geodesic.

Theorem 4.5 (The four-state rotor). *In R with the fixed maximum cut there are exactly four selections,*

$$\omega_{m,x} = \{A_m, B_x\}, \quad \omega_{m,y} = \{A_m, B_y\}, \quad \omega_{v,y} = \{A_v, B_y\}, \quad \omega_{v,x} = \{A_v, B_x\},$$

and the following hold.

- (i) *The state graph on the four selections, with an edge for each detour move, is the four-cycle $\omega_{m,x} \omega_{m,y} \omega_{v,y} \omega_{v,x}$.*
- (ii) *In the state $\omega_{u,z}$ the two selected geodesics intersect exactly in $\{u, z\}$, which spans an edge of the square; each of the four square edges arises as the owner pair of exactly one state. In particular, no selection is disjoint.*
- (iii) *In every state $|S_\omega| = 7$: exactly one crossing edge is unselected, namely the square edge vertex-disjoint from the owner pair.*

Proof. By Lemma 4.3 each bad edge has exactly two geodesics, so the selections are the four listed pairs, and a detour move toggles exactly one of the two coordinates ($A_m \leftrightarrow A_v$ or $B_x \leftrightarrow B_y$). Two states are joined by a move iff they differ in exactly one coordinate, which is the stated four-cycle; this proves (i). Parts (ii) and (iii) are read off from the vertex and edge sets of the geodesics:

state	selected geodesics	owner pair	unselected B -edge
$\omega_{m,x}$	A_m, B_x	$\{m, x\}$	yv
$\omega_{m,y}$	A_m, B_y	$\{m, y\}$	vx
$\omega_{v,y}$	A_v, B_y	$\{v, y\}$	xm
$\omega_{v,x}$	A_v, B_x	$\{v, x\}$	my

For instance $V(A_m) \cap V(B_x) = \{a, x, m, y, b\} \cap \{p, m, x, v, q\} = \{m, x\}$ and $E(A_m) \cup E(B_x) = \{ax, xm, my, yb\} \cup \{pm, mx, xv, vq\} = B \setminus \{yv\}$; the other three rows are identical checks. $\square$

Thus every state has exactly one colliding pair of geodesics, sharing two adjacent vertices, and every detour move merely slides this collision to an adjacent edge of the square. The following automorphism shows that the four states are genuinely indistinguishable: no isomorphism-invariant tie-break distinguishes them.

Proposition 4.6 (Neutrality). *The permutation $\sigma = (xmyv)(apbq)$ is an automorphism of R of order four. It exchanges the two shores, hence preserves the cut, B , and M ; it maps geodesics cyclically,*

$$A_m \mapsto B_y \mapsto A_v \mapsto B_x \mapsto A_m,$$

and it advances the rotor by one step:

$$\sigma \omega_{m,x} = \omega_{m,y}, \quad \sigma \omega_{m,y} = \omega_{v,y}, \quad \sigma \omega_{v,y} = \omega_{v,x}, \quad \sigma \omega_{v,x} = \omega_{m,x}.$$

In particular the four states form a single orbit of a cut-preserving automorphism: the collision pattern rotates around the square and never shrinks or grows.

Proof. Direct check on the ten edges: σ maps ax, yb, pm, vq to pm, vq, yb, ax ; the square edges xm, my, yv, vx to my, yv, vx, xm ; and ab, pq to pq, ab . Also $\sigma(V_0) = \{m, v, b, a\} = V_1$. Applying σ vertexwise to the paths of Lemma 4.3, $\sigma(A_m) = pmyvq = B_y$, $\sigma(B_y) = byvxa = A_v$ (reversed), $\sigma(A_v) = pmxvq = B_x$, and $\sigma(B_x) = bymxa = A_m$ (reversed). The action on states follows by applying σ to each coordinate. $\square$

We call R , with its fixed maximum cut, a *neutral square rotor*: *square* because the owner pair and the unselected crossing edge travel around a fixed four-cycle of B , and *neutral* because an order-four cut-preserving automorphism makes all states of the repair walk equivalent.

Remark 4.7 (What the rotor does and does not show). The rotor is not a Hall violation, let alone a support circuit: a support circuit has support union one edge smaller than its atom set, whereas here $|M| = 2$, $|\text{supp}_B(M)| = 8$, and even each single selected support has seven edges. Moreover the unselected crossing edge of each state has both endpoints on the square, which meets no bad edge. What the example eliminates is any argument that upgrades support abundance to a disjoint system of geodesic representatives, or that assumes collision repair by detour moves terminates: on R all four selections carry the same single collision, and the repair walk never terminates. Triangle-freeness, maximum-cut optimality, and a spare geodesic at every bad edge do not preclude this. Whether a rotor can be embedded in a larger triangle-free graph, keeping the cut maximum, so that the rotating collision couples to a genuine support deficiency, is open; no such graph is known. Finally, $\text{bip}(R) = 2 < 8^2/25$: as with the static obstructions, the example constrains proof strategies and has no bearing on the truth of the Erdős $n^2/25$ conjecture.

Remark 4.8 (Verification). All proofs above are complete and elementary. Independently, every numbered statement of this section is machine-verified by two exhaustive scripts included in the ancillary archive. The first (`_claude_r39_8vtx_rotor_gate.py`, SHA-256 prefix 6d74bcbd) checks triangle-freeness, bipartiteness and connectivity of B ; $\text{MaxCut}(R) = 8$ by enumeration of all 2^8 cuts; completeness of the two geodesic families by exhaustive search over four-edge crossing paths; and, for each of the four states, the selected support, the unselected edge, the owner pair, and the one-coordinate structure of the four detour transitions. The second (`verify_rotor8_automorphism.py`, SHA-256 prefix 756cc65a) checks the automorphism of Proposition 4.6: that σ preserves adjacency on all vertex pairs, has order exactly four, exchanges the shores and fixes B and M setwise, and—after recomputing $d_B = 4$ by breadth-first search and re-enumerating the geodesic families and the four selections from scratch—maps the geodesics and the states in the stated four-cycles, in particular acting transitively on the states.

5 No balanced rotor on nine-atom circuits ($t = 3$)

Section 4 exhibited a two-atom configuration, far from any Hall violation, on which the local repair walk rotates forever. This section treats the corresponding question on an actual support circuit. The motivation is a defect-transport strategy for excluding support circuits: one tries to show that some quantity strictly decreases along detour moves, and the terminal objects such a strategy must exclude are cyclic families of detour moves along which a locally deficient vertex configuration merely rotates — here with at most three owners, the engine-relevant range on nine atoms (Remark 5.4). We call these *balanced rotors* and define them precisely below.

In the window parametrization of the transport analysis, where a circuit of t^2 atoms carries rotating vertices with t crossing and t monochromatic neighbours, the smallest case is $t = 3$: nine atoms and, by the circuit identity, eight support edges. We give two independent proofs that no balanced rotor exists on a nine-atom support circuit. The first counts support edges and is insensitive to the shape of the circuit; the second shows that on the unique nine-atom footprint the coverage condition that any rotation needs is unsatisfiable. Both have kernel-checked Lean counterparts (Remark 5.11).

Throughout the section, G is a finite triangle-free graph with a fixed cut $V(G) = V_0 \dot{\cup} V_1$, B is the crossing subgraph, assumed connected, $M = E(G) \setminus B$, and $(F, \mathcal{A})$ is a support circuit in the graph form of Corollary 2.7: $\mathcal{A} \subseteq \{xy \in M : d_B(x, y) = 4\}$ is inclusion-minimal with $|\text{supp}_B(\mathcal{A})| < |\mathcal{A}|$, and $F = \text{supp}_B(\mathcal{A})$ is viewed as the bipartite subgraph it spans, so that $|E(F)| = |\mathcal{A}| - 1$ and every B -geodesic of an atom lies in F . As in Definition 4.4, a *selection* ω assigns to each atom one of its B -geodesics; $S_\omega \subseteq E(F)$ denotes the union of the selected edge sets, and a *detour move* replaces the selected geodesic of a single atom by a different geodesic of the same atom. Maximality of the cut is not used in this section, except in the motivational Remark 5.4.

5.1 Rotating owners and balanced rotors

Definition 5.1 (Rotating owner profile; covered star). Let ω be a selection. A vertex v has the *rotating owner profile* at ω if:

- (P1) $d_B(v) = d_M(v) = 3$, and the three monochromatic edges at v are atoms of $\mathcal{A}$;
- (P2) writing $N_B(v) = \{x, y_1, y_2\}$, exactly one of the three crossing edges at v lies outside S_ω ; label the notation so that this edge is vx (the *active edge*), so $vy_1, vy_2 \in S_\omega$;
- (P3) (*covered star*) for each $i \in \{1, 2\}$ some selected geodesic contains both x and y_i but not v .

Remark 5.2. The definition keeps only the conditions used in the proofs below. In the transport engine, a rotor state satisfies strictly stronger local conditions (prescribed counts of selected geodesics through the owner and two exact matching conditions), and its transitions carry an exactly balanced deficiency ledger, which is the origin of the name *balanced* in Definition 5.3. Excluding the weaker object above therefore suffices. Requiring in (P3) that the covering geodesic avoid v also costs nothing: if a selected geodesic $w_0w_1w_2w_3w_4$ contained both v and x , then, positions of v and x having opposite parity because $vx \in B$, either they are adjacent on the geodesic, making vx a selected edge and contradicting (P2), or they sit at positions $\{0, 3\}$ or $\{1, 4\}$, and the chord vx yields a B -path of length two between the atom endpoints, contradicting $d_B(w_0, w_4) = 4$.

Definition 5.3 (Balanced rotor). A *balanced rotor* on $(F, \mathcal{A})$ consists of $k \in \{2, 3\}$ distinct vertices $v_1, \dots, v_k$ (the *rotating owners*) together with selections $\omega_1, \dots, \omega_k$ such that, with indices modulo k :

- (R1) v_i has the rotating owner profile at ω_i ;
- (R2) ω_{i+1} is obtained from ω_i by a single detour move replacing a selected geodesic of the form $axv_{i+1}yb$ by the geodesic axv_iyb of the same atom $ab \in \mathcal{A}$.

Condition (R2) says that consecutive owners occur as the two middle vertices of a pair of geodesics of one atom that agree elsewhere; this is exactly the one-atom detour move of the repair walk, as on the four-state rotor of Section 4.

Remark 5.4 (Why two or three owners). A nontrivial rotation needs at least two owners, and the engine's four-owner configuration is excluded on nine atoms by a separate count, which we

record for orientation; it is the only place in this section where maximality of the cut enters. In that configuration the four rotating vertices are the corners of a crossing four-cycle, each is cut-tight in the sense of Definition 3.1, and each has at least four crossing neighbours. The four monochromatic stars are pairwise disjoint: adjacent corners are joined by crossing edges, while opposite corners lie on one shore and have common crossing neighbours, so an edge between them would close a triangle. Corollary 3.5 gives $d_M \geq d_B - 1 \geq 3$ at each corner, whence $|M| \geq 12 > 9$. The finite core of this count is kernel-checked (Remark 5.11). Definition 5.3 therefore fixes $k \in \{2, 3\}$.

5.2 First closure: the support-edge count

Lemma 5.5 (Owner degree in the support graph). *Let ω be a selection and let v have the rotating owner profile at ω . Then all three crossing edges vx, vy_1, vy_2 are edges of F ; in particular $v \in V(F)$ and $\deg_F(v) \geq 3$.*

Proof. The edges vy_1, vy_2 lie in $S_\omega \subseteq E(F)$ by (P2). For vx , take the covering geodesic of (P3) with $i = 1$: a selected geodesic $Q = w_0w_1w_2w_3w_4$ of some atom $ab \in \mathcal{A}$ with $x, y_1 \in V(Q)$ and $v \notin V(Q)$. The vertices $x, y_1 \in N_B(v)$ lie on the shore opposite v , and the positions of Q alternate shores, with $w_0 = a$ and $w_4 = b$ on one shore. If a, b lie on the shore of v , then x and y_1 occupy the two odd positions w_1, w_3 . If a, b lie opposite v , then x, y_1 occupy even positions; they cannot be $\{w_0, w_4\}$, since then $xy_1 = ab \in M$ and $vx, vy_1 \in B$ would form a triangle. In every case x and y_1 occur at distance two along Q , separated by a middle vertex w , and $w \neq v$ because $v \notin V(Q)$. Replacing w by v in Q yields a walk from a to b of length four whose two new edges vx, vy_1 lie in B and whose vertices are distinct; it is therefore a B -geodesic of ab , since $d_B(a, b) = 4$. Its edges lie in $\text{supp}_B(ab) \subseteq E(F)$ by the definition of the support. Hence $vx \in E(F)$. The three edges are distinct because x, y_1, y_2 are. $\square$

Lemma 5.6 (One shore, pairwise nonadjacent). *Let $v_1, \dots, v_k, \omega_1, \dots, \omega_k$ be a balanced rotor on $(F, \mathcal{A})$. Then all rotating owners lie on one shore, any two of them are nonadjacent in G , and no edge of F is incident to two owners. In particular, the sets of F -edges incident to the several owners are pairwise disjoint.*

Proof. Consider the move (R2) from ω_i to ω_{i+1} , at an atom ab . Both geodesics $axv_{i+1}yb$ and axv_iyb are five-vertex alternating paths whose endpoints a, b , being the endpoints of a monochromatic edge, lie on one shore; the middle position therefore lies on the endpoint shore as well. Hence v_i and v_{i+1} lie on a common shore, and around the cycle of moves all owners lie on one shore. Consequently $v_i v_{i+1} \notin B$; and if $v_i v_{i+1} \in M$, then, since $xv_i, xv_{i+1} \in B$ by the two displayed geodesics, the vertices x, v_i, v_{i+1} would form a triangle in G . So consecutive owners are nonadjacent. For $k \leq 3$ every pair of owners is consecutive in the cyclic order, so no two owners are adjacent, and an edge of G incident to two owners would be exactly such an edge. $\square$

Theorem 5.7 (First closure: support-edge counting). *A nine-atom support circuit carries no balanced rotor.*

Proof. Suppose $(F, \mathcal{A})$ with $|\mathcal{A}| = 9$ carries a balanced rotor; by Corollary 2.7(a), $|E(F)| = 8$. Each owner v_i has at least three incident edges in F (Lemma 5.5, applied at ω_i ; note that $E(F)$ does not depend on the selection), and the incident sets of distinct owners are disjoint (Lemma 5.6).

If $k = 3$, the three disjoint triples already give $|E(F)| \geq 9 > 8$, a contradiction.

If $k = 2$, write $v = v_1, m = v_2$. The two incident triples give six edges of F , and we exhibit three further edges of F incident to neither owner. By (P1) at v , the three monochromatic edges vb_1, vb_2, vb_3 at v are atoms; the vertices b_1, b_2, b_3 are distinct, lie on the shore of v , and none equals m , since $vm \notin E(G)$ by Lemma 5.6 while $vb_i \in M$. By Corollary 2.7(c), $d_F(v, b_i) = 4$; choose an F -geodesic P_i from v to b_i for each i . Two cases.

Some v - b_i geodesic in F avoids m . Choose P_i to be such a geodesic. Its last three edges are three distinct edges of F ; none is incident to v , which occurs only at the first position of the path, and none is incident to m , which does not occur at all.

Every v - b_i geodesic in F contains m , for each i . Fix any choice of $P_i = v p_i h_i q_i b_i$. Since m lies on the shore of v , it occupies an even position of P_i ; it is neither position 0 (which is $v \neq m$) nor position 4 (which is $b_i \neq m$), so $m = h_i$ for every i . The three final edges $q_i b_i$ then avoid both owners: q_i , at an odd position, lies opposite the shore of v and m , and $b_i \notin \{v, m\}$. They are pairwise distinct because each has exactly one endpoint, namely b_i , on the shore of v , and the b_i are distinct.

In either case $E(F)$ contains three edges of F meeting neither v nor m , disjoint from the six incident edges. Hence $|E(F)| \geq 3 + 3 + 3 = 9 > 8$, a contradiction. $\square$

5.3 Second closure: coverage emptiness on the nine-atom footprint

The second proof does not count support edges. It shows that on the unique nine-atom footprint the covered-star clause (P3) can never be satisfied, at any vertex and any selection. The row-level statement is general.

Theorem 5.8 (Row exclusion at a closed star). *Let $(F, \mathcal{A})$ be a support circuit and let z be a vertex such that every atom of $\mathcal{A}$ has both endpoints on the shore of z , and every atom has at least one endpoint in $\{z\} \cup N_M(z)$, where $N_M(z)$ is the set of monochromatic neighbours of z . Let ω be a selection and let $x, s \in N_B(z)$ be distinct with $zx \notin S_\omega$. Then no selected geodesic of ω contains both x and s .*

Proof. Suppose the selected geodesic $Q = a q_1 q_2 q_3 b$ of an atom $ab \in \mathcal{A}$ contains both x and s . The endpoints a, b lie on the shore of z by hypothesis, so the even positions a, q_2, b of Q lie on the shore of z , while $x, s \in N_B(z)$ lie on the opposite shore. Hence $\{x, s\} = \{q_1, q_3\}$. By the second hypothesis applied to ab , one of the following holds.

$z = a$. If $x = q_1$, then $zx = a q_1 \in E(Q) \subseteq S_\omega$, contradicting $zx \notin S_\omega$. If $x = q_3$, then $zx \in B$ and $a q_3 b$ is a B -path of length two, contradicting $d_B(a, b) = 4$.

$z = b$. Symmetric, exchanging q_1 and q_3 .

$za \in M$. The vertex q_1 is one of x, s , so $z q_1 \in B$; also $a q_1 \in E(Q) \subseteq B$ and $za \in M$. Then z, a, q_1 form a triangle in G , a contradiction.

$zb \in M$. Symmetric, with the triangle z, b, q_3 . $\square$

Corollary 5.9 (Second closure: coverage emptiness). *No vertex of a nine-atom support circuit has the rotating owner profile at any selection. In particular, a nine-atom support circuit carries no balanced rotor.*

Proof. By the exact footprint classification of the companion paper [6], a nine-atom support circuit has, up to isomorphism, the support graph

$$\{l_1, l_2, l_3\} - c_1 - c_2 - c_3 - \{r_1, r_2, r_3\}, \quad \mathcal{A} = \{l_i r_j : 1 \leq i, j \leq 3\},$$

the double star whose atom set is a $K_{3,3}$ on the six vertices l_i, r_j . Every atom is monochromatic, so its endpoints share a shore, and since the atom graph $K_{3,3}$ is connected, all six vertices l_i, r_j lie on one shore.

Suppose a vertex z has the rotating owner profile at a selection ω . By (P1) the three monochromatic edges at z are atoms, so z is an endpoint of atoms; say $z = l_i$ (the case $z = r_j$ is symmetric). Then every atom $l_a r_b$ has the endpoint $r_b \in N_M(z)$, because $l_i r_b \in \mathcal{A} \subseteq M$; and every atom has both endpoints on the shore of z . Thus Theorem 5.8 applies to z : since $zx \notin S_\omega$ by (P2), no selected geodesic contains both x and y_1 . This contradicts (P3). $\square$

Remark 5.10 (Independence of the two closures). Theorem 5.7 uses no classification and applies to every nine-atom circuit shape, whatever its footprint; it does not use coverage emptiness. Corollary 5.9 counts nothing, but consumes the footprint classification of [6], which is computer-assisted with two independent exact verifiers. Either result alone excludes balanced rotors on nine atoms; the second is in one respect stronger, excluding even a single profiled vertex rather than only a cyclic family.

Remark 5.11 (Formalization). Both closures are verified in Lean 4 in the development `Erdos23Delta0` accompanying this project, in the following division of labour. All named declarations compile with axioms exactly `propext`, `Classical.choice`, `Quot.sound`; none uses `sorry` or `native_decide`.

- (i) For Theorem 5.7, the finite counting core is the module `Erdos23Delta0.Gamma.R43SupportIncidence` (source SHA-256 prefix `66649670`). Abstract carriers package an eight-edge support set together with, per owner, a three-element incident subset with explicit distinctness witnesses, the pairwise disjointness facts, and, for two owners, the external triple; the kernel-checked theorems `fullyCoveredLiveStar_fullSupportDegree_ge_three`, `twoRotatingOwners_force_nine_support`, `threeRotatingOwners_force_nine_supportEdges`, and `not3_balancedDeficiencyRotor` derive the contradiction $9 \leq |E(F)| = 8$. The graph-level inputs, namely Lemmas 5.5 and 5.6 and the two-case argument in the proof of Theorem 5.7, are proved on paper above and enter the formal statement as hypotheses of the carriers.
- (ii) For Theorem 5.8, the formalization is at graph level, over the development’s literal edge-list graphs, explicit five-vertex checked rows, and executable pair counts. The row-local statement is `bad_star_cover_row_impossible` in `Erdos23Delta0.Gamma.BadStarCoverFreeness` (SHA-256 prefix `afd944ea`); its endpoint cases use the checked-row inducedness lemma `checkedRow_blue_cooccur_implies_pathEdge` of `Erdos23Delta0.Gamma.SaturatedRotorSupportPers`. The selection-level statement, with the closed-star hypotheses of Theorem 5.8 as a structure and the conclusion that the production pair count of x, s vanishes, is `pairCount_eq_zero_of_closedBadStar` in `Erdos23Delta0.Gamma.K33BadStarPairCountZero` (SHA-256 prefix `35456523`).
- (iii) The disjoint-star count of Remark 5.4 has its finite core in `Erdos23Delta0.Gamma.CutTightActiveRotorIn` (SHA-256 prefix `0814a665`): `twelve_le_ambient_card` and `not_ambient_card_nine`.

Remark 5.12 (Scope). Both theorems close only the nine-atom window. At the next window, sixteen atoms and $|E(F)| = 15$, the analogous counts give two owners of support degree at least $t = 4$ plus an external family of size t , in total $3t = 12 \leq 15$ edges, and no contradiction results; the exclusion of larger windows requires different arguments and is not addressed in this section. As with the static obstructions of the companion paper, these results constrain proof strategies: they yield no upper bound on $\text{bip}(G)$ and have no bearing on the truth of the Erdős $n^2/25$ conjecture.

6 The sixteen-atom closure

The classification of the companion paper [6] stops at ten atoms. This section treats the first structured window beyond it: support circuits with sixteen atoms carrying two *rotating owners* of width four. We first prove, by hand, the crossover bounds that isolate this window and leave it a margin of exactly one support edge; we then close the window by an exhaustive computation with two independent verifiers. The closure eliminates one local repair mechanism for Hall expansion. It does not resolve the Erdős $n^2/25$ conjecture.

6.1 Support circuits

Throughout this section F is a connected bipartite graph with parts X and Y , and $\mathcal{A}$ is a set of m distinct *atoms*: pairs $\{p, q\}$ of vertices of F with $d_F(p, q) = 4$. Both endpoints of an atom lie in one part. A *row* of an atom a is a four-edge F -geodesic between its endpoints, written as a vertex sequence $(p_0, p_1, p_2, p_3, p_4)$; the support $P_F(a)$ is the union of the edge sets of all rows of a . An m -footprint obstruction is defined as in the companion paper: F has $m - 1$ edges, the graph $(V(F), \mathcal{A})$ is triangle-free, the supports cover $E(F)$, and every edge of F lies in at least two supports.

Definition 6.1. A *support circuit* is an m -footprint obstruction $(F, \mathcal{A})$ such that, in addition, for every $a \in \mathcal{A}$ the family $\{P_F(b) : b \in \mathcal{A} \setminus \{a\}\}$ admits a system of distinct representatives that uses every edge of F exactly once.

The name records that $\mathcal{A}$ is a circuit of the transversal matroid of the support incidence system: deleting any one atom restores a perfect matching onto the support edges.

Lemma 6.2 (Transfer). *Let G be triangle-free with a cut (V_0, V_1) , crossing graph B connected, and let $\mathcal{S} \subseteq M$ be inclusion-minimal with $|\text{supp}_B(\mathcal{S})| < |\mathcal{S}| = m$. Put $F = \text{supp}_B(\mathcal{S})$. Then $(F, \mathcal{S})$ is an m -atom support circuit, and its atom-edge incidence graph is connected.*

Proof. The minimal-footprint lemma of the companion paper [6] gives all conditions of an m -footprint obstruction, with every atom at F -distance four and its support equal to the union of its length-four F -geodesics. For the representatives, fix $a \in \mathcal{S}$. By minimality every subfamily of $\mathcal{S} \setminus \{a\}$ satisfies Hall's condition inside F , so by Hall's theorem $\mathcal{S} \setminus \{a\}$ has a system of distinct representatives in $E(F)$; since $|\mathcal{S} \setminus \{a\}| = m - 1 = |E(F)|$, the system uses every edge. For connectivity, split the incidence graph into components with atom classes $\mathcal{A}_i$ and edge classes F_i . Every edge lies in some support, so every component contains an atom, and every support is contained in the component of its atom. If there were at least two components, then from $\sum_i |F_i| = m - 1 < \sum_i |\mathcal{A}_i|$ some component would satisfy $|F_i| < |\mathcal{A}_i|$, and $\mathcal{A}_i$ would be a proper support-deficient subfamily, contradicting minimality. $\square$

Lemma 6.3. *Every row is an induced path of F : if u, w lie on a row R and $uw \in E(F)$, then $uw \in E(R)$.*

Proof. A chord between positions $i < j$ of R , $j - i \geq 2$, yields a walk of length $i + 1 + (4 - j) \leq 3$ between the endpoints of R 's atom, contradicting distance four. $\square$

Definition 6.4. A *selection* σ chooses one row $\sigma(a)$ for each atom $a \in \mathcal{A}$; its *selected support* $\text{supp}(\sigma)$ is the union of the edge sets of the chosen rows. Given σ , the star of a vertex v is σ -covered if there is a neighbour x_0 of v in F (the *active neighbour*) such that $vx_0 \notin \text{supp}(\sigma)$ and, for every other F -neighbour y of v , some chosen row contains both x_0 and y .

Definition 6.5. Let $(F, \mathcal{A})$ be a support circuit. A *rotating family of width t* is a set of k distinct vertices $v_1, \dots, v_k$, all in one part of F , such that every v_i has at least t neighbours in F and is an endpoint of at least t atoms, and every two v_i, v_j have a common F -neighbour. Its members are called *owners*.

The window (t, k) arises from a local strategy that tries to repair failed Hall expansion by re-routing supports around k designated vertices whose stars rotate between selections. Whatever the merits of that strategy, its combinatorial footprint is precise: a t^2 -atom support circuit carrying a rotating family of k owners of width t whose stars are covered by a common selection. The statements below concern only this footprint.

6.2 Crossover bounds

Theorem 6.6. *Let $(F, \mathcal{A})$ be an m -atom support circuit carrying a rotating family $v_1, \dots, v_k$ of width $t \geq 2$.*

- (i) $|E(F)| \geq kt + t$; equivalently $m \geq kt + t + 1$.
- (ii) If $k = 2$ and the star of v_1 is σ -covered for some selection σ , then $|E(F)| \geq 3t + 2$; equivalently $m \geq 3t + 3$.

Proof. Say the owners lie in X .

(i) Distinct vertices of one part are never adjacent in F , so the F -stars of the owners are pairwise disjoint edge sets, and they contribute at least kt edges. Now fix $v = v_1$ and distinct atom-partners $b_1, \dots, b_t$ of v ; each satisfies $d_F(v, b_j) = 4$, so b_j is not an owner: an owner $v_l \neq v$ has a common neighbour with v and hence $d_F(v, v_l) = 2$. For each j choose a row R_j of the atom $\{v, b_j\}$ and let f_j be its terminal edge at b_j , with endpoints $b_j \in X$ and $q_j \in Y$. The f_j are pairwise distinct, since their X -endpoints b_j are distinct; and none is incident with an owner, since $q_j \in Y$ while the owners lie in X , and b_j is not an owner. This gives t further edges, hence $|E(F)| \geq kt + t$, and $m - 1 = |E(F)|$ finishes (i).

(ii) Keep the $2t$ owner-incident edges and the t terminal edges from (i), taken at $v = v_1$. Let x_0 be the active neighbour and let $y \neq x_0$ be any other F -neighbour of v (it exists since $t \geq 2$). By coverage some chosen row Q contains x_0 and y . First, $v \notin Q$: otherwise v and x_0 both lie on the induced path Q (Lemma 6.3) and $vx_0 \in E(F)$, so vx_0 would be an edge of $Q \subseteq \text{supp}(\sigma)$, contradicting the choice of x_0 . Hence no edge of Q is incident with v , and at most two of the four edges of Q are incident with v_2 , so at least two edges of Q are incident with no owner.

It remains to show that no edge of Q equals a terminal edge f_j . Suppose $f_j \in E(Q)$; then $b_j \in V(Q)$. The vertices x_0, y are F -neighbours of $v \in X$, so $x_0, y \in Y$, while $b_j \in X$. If the atom of Q lies in X , the Y -vertices of Q are its two internal odd positions, so $\{x_0, y\}$ is exactly that pair, and every X -vertex of Q — in particular b_j — has all its Q -neighbours in $\{x_0, y\}$. If the atom of Q lies in Y , then b_j occupies an internal odd position and its two Q -neighbours lie in the three Y -positions, which include x_0 and y ; by pigeonhole b_j is again Q -adjacent to x_0 or y . Either way F contains an edge from b_j to a neighbour of v , so $d_F(v, b_j) \leq 2$, contradicting $d_F(v, b_j) = 4$. Thus the two owner-free edges of Q are new, and $|E(F)| \geq 2t + t + 2 = 3t + 2$. $\square$

Corollary 6.7 (Crossover table). *Let $(F, \mathcal{A})$ be a support circuit with $m = t^2$ atoms, $t \geq 3$, carrying a rotating family of $k \geq 2$ owners of width t whose stars include one covered star when $k = 2$. Then:*

- (a) $k \geq t - 1$ is impossible; in particular no such family exists for $t = 3$;
- (b) for $k = 2$, necessarily $t \geq 4$;
- (c) at $t = 4$ the only surviving pair is $(t, k) = (4, 2)$, and its margin is a single edge: the bound $3t + 2 = 14$ against the budget $t^2 - 1 = 15$.

Proof. (a) If $k \geq t - 1$ then $kt + t \geq t^2 > t^2 - 1 = |E(F)|$, contradicting Theorem 6.6(i); for $t = 3$ this excludes every $k \geq 2$. (b) $3t + 2 \leq t^2 - 1$ fails for $t = 3$ ($11 > 8$) and holds first at $t = 4$ ($14 \leq 15$). (c) At $t = 4$, part (a) kills $k \geq 3$ ($kt + t \geq 16 > 15$). $\square$

6.3 The two-owner window at $t = 4$

Definition 6.8. A *two-owner window* is a triple $(F, \mathcal{A}, \{v, m\})$, where $(F, \mathcal{A})$ is a 16-atom support circuit and $v \neq m$ are vertices in one part of F such that $\deg_F(v) = \deg_F(m) = 4$, v and m have at least two common F -neighbours, and each of v, m is an endpoint of exactly four atoms. The window is *covered* if a single selection σ -covers the stars of both owners.

t	budget $t^2 - 1$	$k = 2$ bound $3t + 2$	$k = 3$ bound $4t$
3	8	11 (excluded)	12 (excluded)
4	15	14 (margin 1)	16 (excluded)
5	24	17 (margin 7)	20 (margin 4)
6	35	20 (margin 15)	24 (margin 11)

Table 1: The crossover bounds of Theorem 6.6 against the edge budget of a t^2 -atom support circuit. All families with $k \geq t - 1$ are excluded outright. The first surviving window is $(t, k) = (4, 2)$, with margin exactly one.

The exact degree and atom counts in Definition 6.8 reflect the graph-level states behind the window, in which each owner has exactly four crossing and four monochromatic neighbours. That the *full* crossing star of a covered owner is forced into the support union — so that $\deg_F = 4$ rather than $\deg_F \leq 4$ — is the content of the next lemma.

Lemma 6.9 (Star absorption). *In the setting of Lemma 6.2, let v be a vertex with exactly four B -neighbours, let σ select one B -geodesic row for each atom of $\mathcal{S}$, and suppose there is a B -neighbour x_0 of v with $vx_0 \notin \text{supp}(\sigma)$ such that each pair $\{x_0, y\}$, for the other three B -neighbours y of v , lies on a common chosen row. Then all four B -star edges of v belong to $F = \text{supp}_B(\mathcal{S})$, and $\deg_F(v) = 4$.*

Proof. Fix y and a chosen row Q containing x_0 and y . As in Theorem 6.6(ii), $v \notin Q$: rows are induced (Lemma 6.3 applies verbatim to B -geodesics), and $v \in Q$ would force the non-selected edge vx_0 into Q . The vertices x_0, y lie in the shore opposite v , hence at even distance along Q ; distance four would make $\{x_0, y\}$ the atom of Q , impossible because $d_B(x_0, y) = 2$ through v . So x_0, y lie at Q -distance two, with a middle vertex $q \neq v$. Replacing q by v turns Q into another four-edge B -path with the same endpoints — a path because $v \notin Q$ — hence another row of the same atom. Its edges vx_0 and vy therefore lie in the complete support of that atom, which is contained in F . $\square$

Proposition 6.10 (The $K_{2,3}$ core). *In every two-owner window $(F, \mathcal{A}, \{v, m\})$ the owners share an atom-partner: there is a vertex b with $\{v, b\}, \{m, b\} \in \mathcal{A}$. Consequently, if x, y are two common neighbours of the owners, the window contains the mixed $K_{2,3}$ core on $\{v, m\} \cup \{x, y, b\}$: the four support edges vx, vy, mx, my and the two atoms $\{v, b\}, \{m, b\}$.*

Proof. The two owner stars are disjoint and contribute eight of the fifteen edges. As in Theorem 6.6(i), each owner has four distinct terminal edges, none incident with an owner; so the eight star edges and the terminal edges of v and of m would number $8 + 4 + 4 = 16 > 15$ if the two terminal families were disjoint. Hence some terminal edge of v equals one of m ; equal edges have equal endpoints in the owners' part, so the corresponding atom-partners coincide. $\square$

The finite incidence core of Proposition 6.10 is kernel-checked in Lean 4 as `K2TailIncidence.exists_common` (module `R44K2TailOverlap.lean`, source SHA-256 prefix 12DFB927), compiling with axioms exactly `propext`, `Classical.choice`, `Quot.sound`. It is a structural reduction, not by itself an exclusion.

Lemma 6.11 (Vertex range). *Every two-owner window satisfies $8 \leq |V(F)| \leq 15$. If some selection covers the star of one owner, then $|V(F)| \leq 14$.*

Proof. Lower bound: $(V(F), \mathcal{A})$ is triangle-free with 16 edges, so $16 \leq |V(F)|^2/4$ by Mantel's theorem [8], giving $|V(F)| \geq 8$. Upper bound: F is connected with 15 edges, so $|V(F)| \leq 16$ with equality only for a tree; but two common neighbours x, y of the owners give the four-cycle $v-x-m-y$, so F is not a tree and $|V(F)| \leq 15$.

Now suppose $|V(F)| = 15$ and the star of v is σ -covered, with active neighbour x_0 and other neighbours y_1, y_2, y_3 . A connected graph with equally many vertices and edges contains exactly one cycle. For each i take a chosen row Q_i containing x_0 and y_i . Exactly as in Lemma 6.9 (whose argument is purely about F -geodesics): $v \notin Q_i$; the pair x_0, y_i cannot lie at Q_i -distance four, since that would make $\{x_0, y_i\}$ the atom of Q_i while v is a common F -neighbour, so $d_F(x_0, y_i) = 2 \neq 4$; hence x_0, y_i lie at Q_i -distance two with middle $q_i \neq v$. Then

$$C_i = v-x_0-q_i-y_i-v$$

is a four-cycle of F , using the star edges vx_0, vy_i and two edges of Q_i . For $i \neq j$ the cycles differ: $y_i \in C_i$, while $y_i \notin \{v, x_0, q_j, y_j\}$ (the middles q_j lie in the owners' part, the y 's in the other). Three distinct cycles contradict unicyclicity, so $|V(F)| \leq 14$. $\square$

6.4 The census

The closure now rests on one exhaustive computation.

Theorem 6.12 (Computer-assisted). *Up to isomorphism there are exactly 153,978 connected bipartite graphs with 15 edges and between 8 and 15 vertices (Table 2). Of these, exactly 34 admit two same-part degree-four vertices v, m with at least two common neighbours, at least four same-part vertices at distance four from each, and a common same-part vertex at distance four from both; each of the 34 admits exactly one such pair. These embeddings extend to exactly 576 two-owner windows $(F, \mathcal{A}, \{v, m\})$, through 74,920 atom completions, of which 2,299 have a triangle-free atom graph, 862 satisfy support coverage and edge multiplicity at least two, and 576 satisfy all sixteen deletion representative systems. Furthermore:*

- (i) *all 576 windows have $|V(F)| = 15$, on exactly four isomorphism types of F ; in particular no two-owner window has at most 14 support vertices;*
- (ii) *in every window, at least eight atoms have every row through v , and at least eight have every row through m ; over the 576 windows the number of such atoms has distribution $\{8:255, 9:193, 10:101, 11:26, 12:1\}$, identically for v and m ;*
- (iii) *no complete row family of any atom of any window contains a middle swap: two rows (a, x, v, y, b) and (a, x, m, y, b) differing precisely in the middle vertex, one through each owner.*

The reduction to a finite computation is complete for two-owner windows: F is connected, bipartite, with fifteen edges and, by Lemma 6.11, between 8 and 15 vertices; the degree, common-neighbour and atom-count conditions are part of Definition 6.8; the four atom-partners of each owner are same-part vertices at distance four; and the common distance-four vertex exists by Proposition 6.10. The computation generated all candidate graphs with **geng** from nauty/Traces [9] (**geng** -c -b n 15:15, $8 \leq n \leq 15$), located the owner embeddings (the filter is invariant under swapping the two colour classes), enumerated all admissible sixteen-atom completions, and tested the circuit conditions with exact integer arithmetic. Every complete row family of every surviving window was then reconstructed and every pair of rows in every family was tested for the middle-swap pattern; the row-selection products of the 576 windows comprise 16,288 selections in total.

Corollary 6.13 (The sixteen-atom closure). *There is no covered two-owner window. Consequently, in a triangle-free graph with a cut whose crossing graph is connected, no inclusion-minimal support-deficient family of sixteen bad edges admits two same-shore vertices of crossing degree four, with at least two common crossing neighbours and exactly four atoms each, whose stars are simultaneously covered by one selection of shortest rows. The window $(t, k) = (4, 2)$ of Table 1 is empty.*

n	graphs	n	graphs
8	2	12	15,285
9	30	13	36,337
10	496	14	52,909
11	3,675	15	45,244

Table 2: Connected bipartite graphs with 15 edges, by vertex count: the census ground set, 153,978 graphs in total. The counts were produced by **geng**, reproduced independently by the NetworkX-based verifier, and replayed once more for this paper.

Proof. Suppose $(F, \mathcal{A}, \{v, m\})$ is a covered two-owner window. By Lemma 6.11 it has at most 14 support vertices; by Theorem 6.12(i) no two-owner window — covered or not — has fewer than 15. For the graph-level statement, transfer by Lemma 6.2: the minimal family induces a 16-atom support circuit whose atoms, rows and supports coincide with the graph-level ones, and Lemma 6.9 forces each covered owner’s full crossing star into F , so $\deg_F(v) = \deg_F(m) = 4$ and the selection covers both stars inside $(F, \mathcal{A})$. $\square$

Two features of Theorem 6.12 make the closure robust. First, parts (ii) and (iii) exclude the window a second time, at $|V(F)| = 15$, independently of the covered-star analysis: the re-routing mechanism that motivates the window both prescribes a routing load of four atoms through each owner — against the forced load of at least eight in (ii) — and requires at least one middle swap in some complete row family, against (iii). Part (iii) is quantified over every pair of rows of every family of every window, with no side conditions. Second, the middle-swap exclusion has a row-free confirmation:

Proposition 6.14. *If two rows of one atom form a middle swap (a, x, v, y, b) , (a, x, m, y, b) , then x, y are distinct common neighbours of v and m , a is a neighbour of x outside $\{v, m\}$, b is a neighbour of y outside $\{v, m\}$, and $\{a, b\}$ is a distance-four pair witnessed by both displayed paths. This implication is kernel-checked in Lean 4 (`LiveMiddleSwapCrossOuter.lean`, theorem `live_middle_swap_has_cross_outer`, source SHA-256 prefix `3DFF7897`; axioms exactly `propext`, `Quot.sound`, attested by both the acceptance record and a fresh rebuild-and-probe at assembly, `anc/lean_axiom_probe/`). Exhausting all ordered choices of (x, y, a, b) over the four support types of the 576 windows (multiplicities 180, 190, 190, 16) yields zero such configurations (artifact SHA-256 prefix `79db75b9`, verdict `PASS_NO_LIVE_MIDDLE_SWAP_GEOMETRY`), so Theorem 6.12(iii) follows again without enumerating rows.*

Remark 6.15 (An independent catalogue and a tight near-miss). A second, independently designed session enumeration attacked the same window by vertex count and reported: no owner distance-four profile at $|V(F)| \in \{10, 11\}$; at most ten same-part distance-four pairs at $|V(F)| = 12$ (short of sixteen); 280 candidate graphs at $|V(F)| = 13$ with no sixteen-atom circuit carrying four owner-star atoms at each owner; and 455 candidate graphs at $|V(F)| = 14$ with no simultaneously covered owner stars. In that route, distance-four pairs within one part are certified by the exact criterion $(Q_L)_{uv} = 0 < (Q_L^2)_{uv}$, where $Q_L = A_H A_H^T$ and A_H is the biadjacency matrix. This catalogue is a single implementation and we use it only as corroboration; the proof of Corollary 6.13 rests on the dual-verified census. The catalogue also exhibited a tight near-miss at $|V(F)| = 14$ (checker SHA-256 prefix `4644e5ab`), whose archived invariants we record: parts $\{0, \dots, 6\}$ and $\{7, \dots, 13\}$; fifteen support edges and sixteen atoms; owners 0 and 1 sharing the two support neighbours 7, 8 and the three atom-partners 3, 5, 6; shared terminal edges $\{3, 13\}, \{5, 13\}, \{6, 13\}$; the remaining edge budget spent on $\{2, 11\}$ and $\{4, 9\}$; a complete row database of 864 selections, none of which covers both owner stars. The recorded failure is local: the common neighbours 7, 8 lie on no geodesic row of the relevant families, and every alternative row passes through an owner and thus absorbs the would-be

active edge. The full edge and atom lists of this configuration were session-internal and are not reproduced here.

Remark 6.16 (Why geometry is load-bearing). The census cannot be replaced by support-cardinality reasoning. There is an abstract sixteen-atom, fifteen-edge transversal circuit with four-element supports — artifact SHA-256 prefix `5b386cd9`, independently re-verified — realizing the forced tail overlap of Proposition 6.10, edge multiplicity at least two, full deficiency one, and Hall’s condition on all 65,534 proper atom subsets. Its supports have not been realized as geodesics of any one graph, and no census window realizes them. Any proof of Corollary 6.13 must therefore use the geodesic geometry, as the census does.

Verification record

The census has two acceptance paths with no shared code beyond the `geng` generator. The primary enumeration (three programs, eight workers) emitted three canonical JSON artifacts with embedded canonical SHA-256 hashes, listed first below; the remaining two JSON artifacts belong to the verification path, and the two Lean sources are hashed alongside:

```
t4.support_graph.census.json
40f16a84559ace4827e366f152026f2b7868bdaed31ff9afb36184a29b48046d
t4.atom_circuit.census.json
302e04ef5ff14c78cbe9dc5800ac0226e730ed0baca123585dc6469a82d66652
t4.profile_transition.census.json
b464682b4142a9db2396dc39ac9a0ffd8ff638aba1b9270734667c8f0a543114
t4.cross_outer_exclusion.json
79db75b95e8401064f1b6159bb980ee0149f0fb3a602a607306a7f0e501a5d49
t4.support_circuit_hit.json (Remark 6.16)
5b386cd90b795bf1e6f8f174e21aa559e37c9f682e5dff373dae6bf74f3b9641
LiveMiddleSwapCrossOuter.lean
3DFF7897F65112F4F8177B84AB28F631C85BC6F50F7F5D920ED06F033B7F9275
R44K2TailOverlap.lean
12DFB92724F36BEAC1F11EBA529023F0445893CA6E60376B09B2403C0E64066A
```

Three independently written verifiers replayed the pipeline: `verify_t4.support_census.py` re-ran `geng` without residue splitting and reproduced, via `networkx.from_graph6_bytes` and `NetworkX` bipartitions and distances, all 153,978 graphs, the 34 owner embeddings, and the complete candidate payload (`PASS_INDEPENDENT_NETWORKX_SUPPORT_CENSUS`); `verify_t4.atom_census.py` reproduced every stage count and the same 576 windows using `networkx.all_shortest_paths` and `NetworkX` bipartite matching (`PASS_INDEPENDENT_NETWORKX_ATOM_CENSUS`); and `verify_t4.profile_exclusion.py` reconstructed every complete row family with its own graph6 decoder and breadth-first search, recomputed both canonical hashes, and confirmed the forced-through histograms of Theorem 6.12(ii) and the absence of raw middle swaps (`PASS_T4_RAW_MIDDLE_SWAP_EXCLUSION`); this last verifier was replayed once more during the audit of the session record, with matching histograms. Neither acceptance path uses floating-point arithmetic; the two paths use different graph6 decoders, shortest-path engines, and matching algorithms. The `geng` counts of Table 2 were additionally re-generated for this paper. The Lean modules named above cover only the two finite implications indicated; the census itself is not kernel-checked, and its formal replay remains open.

Remark 6.17 (Scope). The sixteen-atom closure eliminates a single structured window. It neither improves the general bounds on $\text{bip}(G)$ for triangle-free G nor resolves the Erdos $n^2/25$ conjecture. The neighbouring windows $(t, k) = (5, 2)$ and $(5, 3)$, where Table 1 leaves margins seven and four, are open; the unrooted analogue of the census is infeasible there (the graph count at 24 edges already exceeds 1.9×10^8 at sixteen vertices), so a rooted generation or a structural argument would be required.

7 Partial results at twenty-five atoms, and two certified near-misses

At parameter t the configurations of this paper live in support circuits with t^2 atoms on $t^2 - 1$ support edges. The cases $t = 3$ and $t = 4$ (nine and sixteen atoms) are closed earlier in the paper. This section collects what is known at $t = 5$: twenty-five atoms on twenty-four support edges. The results are partial. We prove two structural theorems by hand (an order window and a selected-support bound), report an exact machine exclusion of support orders fifteen and sixteen, and present two eighteen-vertex circuits that pass every local test and are excluded only by explicit certificates. Orders seventeen through twenty-one remain open, and nothing in this section touches the Erdős $n^2/25$ conjecture itself.

7.1 Support circuits, selections, and owners

Recall the footprint objects. A cut $V(G) = V_0 \dot{\cup} V_1$ of a triangle-free graph has crossing edges B and monochromatic edges M ; each $xy \in M$ with $d_B(x, y) = 4$ carries the support $\text{supp}_B(xy) \subseteq B$ of its shortest geodesics. An inclusion-minimal family of bad edges violating Hall's condition projects, by the footprint lemma of the companion classification paper, to the following abstract object. All bipartite graphs below come with a fixed bipartition into two *classes*.

Definition 7.1. Let $t \geq 3$. A *support circuit at parameter t* is a pair $(F, \mathcal{A})$ where F is a finite connected bipartite graph with $|E(F)| = t^2 - 1$, and $\mathcal{A}$ is a set of t^2 unordered pairs of vertices of F , the *atoms*, such that:

- (a) every atom $\{x, y\}$ joins two vertices of a common class with $d_F(x, y) = 4$; writing $R(a)$ for the set of length-four x - y geodesics of F (the *rows* of a) and $P(a)$ for the union of their edge sets, the sets $P(a)$ cover $E(F)$ and every edge of F lies in at least two of them;
- (b) the *atom graph* $(V(F), \mathcal{A})$ is triangle-free;
- (c) every proper subfamily $\mathcal{A}' \subsetneq \mathcal{A}$ admits distinct representatives: an injection $a \mapsto e_a \in P(a)$.

By Hall's theorem, condition (c) together with $|E(F)| = |\mathcal{A}| - 1$ says exactly that the supports violate Hall's condition with defect one while every proper subfamily satisfies it: support circuits are the inclusion-minimal Hall violations, at $m = t^2$ atoms. At $t = 3$ the unique support circuit is the nine-atom double star of the companion classification [6]. Note that the graphs G_t of [6] are themselves *not* support circuits for $t \geq 4$: their violation has defect far exceeding one, and the nine-atom circuit arises among their minimal subsystems [6].

Definition 7.2. A *selection* ω assigns to each atom a one row $\omega(a) \in R(a)$. The *selected* edge set is $S_\omega = \bigcup_a E(\omega(a))$, and $L_\omega = E(F) \setminus S_\omega$ is the *latent* edge set.

Definition 7.3. Let $(F, \mathcal{A})$ be a support circuit at parameter t , let v be a vertex with $\deg_F(v) = \deg_{\mathcal{A}}(v) = t$, and let $x_0 \in N_F(v)$. A selection ω *realizes a covered rotating owner at (v, x_0)* if

- (i) the only selected rows containing v are the rows of the t atoms at v ;
- (ii) $vx_0 \notin S_\omega$ (the *active* edge);
- (iii) every other edge at v lies in S_ω ;
- (iv) for every $y \in N_F(v) \setminus \{x_0\}$, some selected row contains both x_0 and y .

The pairs $\{x_0, y\}$, $y \in N_F(v) \setminus \{x_0\}$, are the *star pairs*; condition (iv) is the *covered star*.

Definition 7.4. Let ω realize a covered rotating owner at (v, x_0) . The *tail* is the connected component of x_0 in the latent graph $(V(F), L_\omega)$ with the vertex v deleted. The owner is *live* if the tail contains a vertex b with $\{v, b\} \in \mathcal{A}$, or both members of some atom; otherwise it is *dead*.

Definition 7.5. A *rotor core* in $(F, \mathcal{A})$ consists of an atom $\{a, b\}$, two further vertices $v \neq m$ in the class of a , and vertices x, y in the opposite class, such that both (a, x, v, y, b) and (a, x, m, y, b) are rows of $\{a, b\}$. A *balanced rotor at parameter t* is a support circuit at parameter t carrying a rotor core whose middle vertices v, m both satisfy $\deg_F = \deg_{\mathcal{A}} = t$, at least one of which admits a selection realizing a *live* covered rotating owner.

The full rotor object of the programme carries additional bookkeeping on top of Definition 7.5; every such object contains the data listed here, so all exclusions below apply to it a fortiori.

We record four elementary lemmas. Throughout, $(F, \mathcal{A})$ is a support circuit at parameter t and v, x_0 are as in Definition 7.3.

Lemma 7.6 (Tightness). *A selection satisfying (i) exists if and only if every atom not at v has a row avoiding v . If some atom not at v has all of its rows through v , then every selection has at least $t + 1$ selected rows through v .*

Proof. Every row of an atom at v is a geodesic from v , so it contains v ; this gives t selected rows through v under any selection. If every atom not at v has a v -avoiding row, choose such rows for all of them; (i) holds. Conversely an atom not at v all of whose rows contain v contributes a $(t + 1)$ -st selected row through v under every selection. $\square$

Lemma 7.7 (Positions). *Let ω satisfy (i) and (ii). Then no selected row of an atom at v contains x_0 , and every selected row containing x_0 avoids v .*

Proof. A row of an atom at v is a geodesic (v, q_1, q_2, q_3, q_4) , so $d_F(v, q_k) = k$. The vertex x_0 lies in the class opposite v , hence could only be q_1 or q_3 . As q_1 it would put the edge vx_0 into S_ω , against (ii); as q_3 it would force $d_F(v, x_0) = 3$, against $vx_0 \in E(F)$. For the second statement, a selected row through v is a row of an atom at v by (i), and therefore omits x_0 . $\square$

Lemma 7.8 (Step surjection). *Let ω satisfy (i)–(iii). Then the selected row of each atom at v has v as an endpoint and its first edge is vy for some $y \neq x_0$; the resulting map from the t atoms at v to the $t - 1$ edges vy is surjective, taking one value twice and every other value once.*

Proof. A row of an atom at v contains v exactly once, as an endpoint, so its unique edge at v is its first edge, and (ii) rules out vx_0 . Conversely each edge vy , $y \neq x_0$, is selected by (iii); a selected row using vy contains v , is a row of an atom at v by (i), and has vy as its first edge. So the map is onto, and t atoms mapping onto $t - 1$ edges with every fibre nonempty forces fibre sizes $(2, 1, \dots, 1)$. $\square$

Lemma 7.9 (Parity). *Let Q be a row of some atom with $v \notin Q$. Then Q contains at most one star pair.*

Proof. Two star pairs contained in Q would give three distinct vertices of $N_F(v)$ on Q . All of $N_F(v)$ lies in the class opposite v , and the vertices of the path $Q = (q_0, \dots, q_4)$ alternate classes, so members of $N_F(v)$ occupy positions of a single parity; three of them must occupy positions 0, 2, 4. Then $q_0, q_4 \in N_F(v)$ are the endpoints of the atom of Q , and $q_0 v q_4$ is a path in F , so $d_F(q_0, q_4) \leq 2$, contradicting the atom distance four. $\square$

Corollary 7.10 (Coverage; order upper bound). *Let ω realize a covered rotating owner at (v, x_0) , with $t = 5$. Then there are four pairwise distinct atoms $a_1, \dots, a_4$, none at v , whose selected rows $Q_i = \omega(a_i)$ satisfy $v \notin Q_i$ and $\{x_0, y_i\} \subseteq Q_i$, one for each star pair. On Q_i the vertices x_0 and y_i have a common neighbour q_i , and the four 4-cycles $v x_0 q_i y_i$ are linearly independent in the cycle space of F . Consequently the cycle rank of F is at least four and*

$$|V(F)| \leq 24 - 4 + 1 = 21.$$

Proof. By (iv) each star pair lies on a selected row, which avoids v by Lemma 7.7; a row avoiding v is not a row of an atom at v . Each atom contributes one selected row, and by Lemma 7.9 that row contains at most one star pair, so the four star pairs are covered by four distinct atoms. On Q_i the vertices x_0, y_i occupy positions of equal parity; positions $\{0, 4\}$ are impossible (the endpoints of Q_i form the atom a_i , and two neighbours of v are at distance at most two), so the positions differ by two and there is a common row-neighbour $q_i \neq v$. The cycle $C_i = v x_0 q_i y_i$ uses the edges $vx_0, x_0q_i, q_iy_i, y_iv$ of F . Each C_i contains vy_i , while C_j for $j \neq i$ meets v only in vx_0, vy_j ; so each cycle owns a private edge and the four are independent. For a connected graph the cycle rank is $|E| - |V| + 1$. $\square$

7.2 The order window $15 \leq |V(F)| \leq 21$

Fix $t = 5$, a support circuit $(F, \mathcal{A})$, and a selection ω realizing a covered rotating owner at (v, x_0) . Write

$$K = \{v\} \cup N_F(v) \cup N_{\mathcal{A}}(v), \quad |K| = 11,$$

where $N_{\mathcal{A}}(v)$ is the set of five atom-partners of v (they lie in the class of v and are distinct from v ; $N_F(v)$ lies in the opposite class). Vertices outside K are *external*.

Lemma 7.11 (Endpoint trichotomy). *Let Q be a covering row for the star pair $\{x_0, y\}$, that is, a row of some atom with $v \notin Q \supseteq \{x_0, y\}$. Then every vertex of Q in the class of v that is adjacent on Q to a member of $N_F(v)$ is external, and exactly one of the following holds.*

- (I) x_0 and y are interior on Q ; then the atom of Q joins two external vertices of the class of v , and Q contains three external vertices of that class.
- (II) One of x_0, y is an endpoint of Q ; then the other endpoint of Q is an external vertex of the class opposite v , and Q contains two external vertices of the class of v .

Proof. A vertex w of Q in the class of v , adjacent on Q to some $z \in N_F(v)$, satisfies $d_F(v, w) \leq 2$; it is not v because $v \notin Q$, and not in $N_{\mathcal{A}}(v)$ because atoms have distance four. So w is external. As in Lemma 7.9, x_0 and y occupy positions of equal parity on $Q = (q_0, \dots, q_4)$, and not $\{0, 4\}$; so the positions are $\{1, 3\}$ (case I) or $\{0, 2\}, \{2, 4\}$ (case II).

Case I: positions $0, 2, 4$ lie in the class of v , and each of q_0, q_2, q_4 is adjacent on Q to q_1 or q_3 , i.e. to a member of $N_F(v)$; all three are external, and the atom of Q is $\{q_0, q_4\}$.

Case II, say positions $\{0, 2\}$: now q_1, q_3 lie in the class of v and each is adjacent on Q to q_0 or q_2 , members of $N_F(v)$; both are external. The atom of Q is $\{q_0, q_4\}$ with $q_0 \in \{x_0, y\} \subseteq N_F(v)$. If $q_4 \in N_F(v)$ then $d_F(q_0, q_4) \leq 2$, against the atom distance; and q_4 lies in the class opposite v , whose part of K is exactly $N_F(v)$. So q_4 is external. $\square$

Theorem 7.12 (Order lower bound). *Let $(F, \mathcal{A})$ be a support circuit at $t = 5$ admitting a selection that realizes a covered rotating owner. Then $|V(F)| \geq 15$. Together with Corollary 7.10,*

$$15 \leq |V(F)| \leq 21.$$

Proof. By Corollary 7.10 there are four covering rows belonging to four distinct atoms. By Lemma 7.11 each covering row contributes at least two external vertices in the class of v ; a row of type I contributes three, and if all four rows are of type II, their atoms have external endpoints in the opposite class, so at least one external vertex lies there. In either case there are at least three external vertices and $|V(F)| \geq 11 + 3 = 14$.

Suppose $|V(F)| = 14$, so there are exactly three external vertices.

Case 1: some covering row has type I. Its three external vertices lie in the class of v and exhaust the external supply, so no external vertex lies in the opposite class, and by Lemma 7.11 no covering row has type II. Hence all four covering atoms have type I: four *distinct* atoms, each

a pair of external vertices of the class of v . But only three such vertices exist, admitting at most $\binom{3}{2} = 3$ pairs. Contradiction.

Case 2: all four covering rows have type II. Then at least two external vertices lie in the class of v and at least one in the opposite class; with three externals in total this forces exactly $2 + 1$, and the classes of F have sizes

$$\underbrace{1 + 5 + 2}_{\text{class of } v} = 8, \quad \underbrace{5 + 1}_{\text{opposite class}} = 6.$$

Atoms join vertices of a common class and the atom graph is triangle-free, so by Mantel's theorem

$$25 = |\mathcal{A}| \leq \left\lfloor \frac{8^2}{4} \right\rfloor + \left\lfloor \frac{6^2}{4} \right\rfloor = 16 + 9 = 25.$$

Equality forces the atom graph induced on the class of v to be the unique extremal graph $K_{4,4}$, in which every vertex has atom-degree four. But $\deg_{\mathcal{A}}(v) = 5$. Contradiction. $\square$

7.3 The selected-support bound

Theorem 7.13. *Let $t \geq 3$, let $(F, \mathcal{A})$ be a support circuit at parameter t , and let ω realize a covered rotating owner at (v, x_0) . Then*

$$|S_\omega| \geq 3t - 1, \quad \text{hence} \quad |L_\omega| \leq (t^2 - 1) - (3t - 1) = t(t - 3).$$

Proof. Write $Y = N_F(v) \setminus \{x_0\}$, $|Y| = t - 1$, and let $b_1, \dots, b_t$ be the atom-partners of v . We exhibit $3t - 1$ distinct selected edges.

Star edges. The $t - 1$ edges vy , $y \in Y$, are selected by (iii).

Second edges. For each $y \in Y$ pick, by Lemma 7.8, a selected row ρ^y of an atom at v with first edge vy , and let $\sigma(y)$ be its second edge, joining y to a vertex of the class of v . Distinct y give distinct edges $\sigma(y)$: they have distinct endpoints in the class opposite v . This yields $t - 1$ selected edges not incident with v (a row of an atom at v meets v only in its first edge).

Last edges. For each j let $\tau(b_j)$ be the last edge of the selected row of the atom $\{v, b_j\}$, joining b_j to a vertex q_j of the opposite class. The b_j are pairwise distinct, so these are t distinct selected edges, again not incident with v .

Distinctness. The star edges meet v ; the others do not. If $\sigma(y) = \tau(b_j)$ for some y, j , then matching endpoints by class gives $q_j = y \in N_F(v)$ and b_j adjacent to y , whence $d_F(v, b_j) \leq 2$, contradicting the atom distance four. So the three families are pairwise disjoint: $3t - 2$ selected edges so far. None of them contains x_0 : the star edges have endpoints v and $y \neq x_0$; each $\sigma(y)$ joins $y \neq x_0$ to the class of v ; each $\tau(b_j)$ joins b_j to q_j , and $q_j \neq x_0$ because rows of atoms at v avoid x_0 (Lemma 7.7).

One more edge at x_0 . By (iv) some selected row P contains x_0 (any covered star pair supplies one); by Lemma 7.7 $v \notin P$, so P has an edge at x_0 different from vx_0 . It is selected, contains x_0 , and therefore differs from all $3t - 2$ edges above. Hence $|S_\omega| \geq 3t - 1$, and the latent bound follows from $|E(F)| = t^2 - 1$. $\square$

Corollary 7.14. *For $t = 5$: at least 14 of the 24 support edges are selected, at most 10 are latent, and, since $vx_0 \in L_\omega$ while every other edge at v is selected, the tail of Definition 7.4 spans at most 9 latent edges.*

Remark 7.15. At $t = 3$ Theorem 7.13 gives $|L_\omega| \leq 0$, while (ii) forces $vx_0 \in L_\omega$. So no covered rotating owner exists at nine atoms at all — consistent with the unique nine-atom circuit, in which no vertex has support- and atom-degree both equal to three. At $t = 4$ the latent budget is 4; at $t = 5$ it is 10. The budget $t(t - 3)$ grows quadratically, so the tail mechanism exploited below is specific to small t .

Remark 7.16. Lemmas 7.6–7.9 convert the existence of a covered rotating owner at (v, x_0) into a finite test on the row database of the circuit: every atom off v needs a v -avoiding row; every atom at v needs a row whose first edge avoids x_0 ; the atom-to-first-edge relation needs a matching covering the $t - 1$ non-active star edges; and, by Lemma 7.9, the covered star needs $t - 1$ *distinct* atoms, one covering row each, so the pair-to-atom coverage relation needs a matching saturating the star pairs. Conversely, choosing rows along such matchings (arbitrary v -avoiding rows elsewhere) realizes the owner, so the four conditions are exactly equivalent to realizability. This four-part test is what the enumeration below implements.

7.4 Exact exclusion of support orders fifteen and sixteen

The machine layer works with a relaxation. A balanced rotor (Definition 7.5) whose support has n vertices, with p of them in the class of the owners, satisfies the following finite constraint system $(R_{p,n-p})$:

- (R1) F is connected and bipartite with classes of sizes p and $n - p$ and $|E(F)| = 24$;
- (R2) there are vertices v, m, a, b in the first class and x, y in the second with (a, x, v, y, b) and (a, x, m, y, b) paths in F ;
- (R3) $\deg_F(v) = \deg_F(m) = 5$;
- (R4) every vertex of F has positive degree;
- (R5) $d_F(a, b) = 4$;
- (R6) v and m each have at least five vertices of their own class at distance exactly four;
- (R7) at least 25 same-class pairs of $V(F)$ are at distance exactly four.

Indeed (R2), (R3) restate the rotor core and owner degrees, (R5) holds because $\{a, b\}$ is an atom, (R6) because $\deg_{\mathcal{A}}(v) = \deg_{\mathcal{A}}(m) = 5$ and atoms are same-class distance-four pairs, and (R7) because the 25 atoms are distinct such pairs.

Lemma 7.17. *In a balanced rotor at $t = 5$, the class of the owners has at least seven vertices and the opposite class at least five.*

Proof. The owner class contains v, m , and the five atom-partners of v ; these are seven distinct vertices, since $d_F(v, m) = 2$ (via x) while atom-partners of v are at distance four. The opposite class contains $N_F(v)$, of size five. $\square$

So at order n the owner-class size p ranges over $7 \leq p \leq n - 5$: four splits at $n = 15$, five at $n = 16$, six at $n = 17$.

Proposition 7.18 (Computer-assisted). *For $n = 15$ and every split $(p, n - p) \in \{(7, 8), (8, 7), (9, 6), (10, 5)\}$, and for $n = 16$ and every split $(p, n - p) \in \{(7, 9), (8, 8), (9, 7), (10, 6), (11, 5)\}$, the constraint system $(R_{p,n-p})$ is infeasible. Consequently no balanced rotor at $t = 5$ has a support of order 15 or 16, and with Theorem 7.12 the support order of a balanced rotor at $t = 5$ lies in $\{17, \dots, 21\}$.*

The nine infeasibility certificates were produced by the Boolean constraint model `rooted_t5_support_cp_sat`. (OR-Tools CP-SAT; integer and Boolean arithmetic only; the symmetry breaking sorts the degrees of the non-distinguished vertices, which any solution can be relabelled to satisfy). Each verdict is a completed infeasibility proof, not a time-out. Artifact SHA-256 prefixes: `14ac9d93, 8d194613, 66e0813c, 5721cbc5` (order 15, splits as listed) and `8dff49ea, bd0c0320, cc5462d6, b5037e11, 9f9401d0` (order 16). Unlike the certificates of the next subsection, these nine rest on a single solver and have not been independently replayed; we flag this honestly.

7.5 Two certified near-misses at order eighteen

The enumeration at order eighteen produced two support graphs whose circuits pass *every* local test of this section. We print them in graph6 form; vertices 0–8 form the owner class and 9–17 the opposite class, with $v = 0$, $m = 1$, $a = 2$, $b = 3$, $x = 9$ in the labelling of (R2):

```
circuit #298 : Q??????wE_[?EGs?D_@A?C_B???
circuit #264 : Q??????wE_Bws?s?DCD??@?@???
```

The following facts are immediate from the printed strings: each graph has 18 vertices, 24 edges, classes of sizes $9 + 9$, is connected and bipartite, satisfies $\deg_F(0) = \deg_F(1) = 5$ and $d_F(2, 3) = 4$, and carries the rotor core of (R2). The same-class distance-four supply is 32 pairs in #298 and 30 pairs in #264, from which the twenty-five atoms are drawn.

Proposition 7.19 (Computer-assisted). *For each of the two graphs above there is a set $\mathcal{A}$ of twenty-five of its same-class distance-four pairs and a selection ω such that $(F, \mathcal{A})$ is a support circuit at $t = 5$ — in particular the atom graph is triangle-free — and ω realizes a covered rotating owner at (v, x_0) , with $v = 0$ and $x_0 = 17$ for #298, and $v = 0$ and $x_0 = x = 9$ for #264. In #264 the active edge is the core edge vx itself.*

The primary search and an independent exact replay (`verify_t5_local_classifier_hit.py`, exact graph operations and integer bipartite matching, no floating point) agree. Artifact prefixes: `c1d474d7` (#298 hit) with verification artifact `48ce1638`, and `6595501f` (#264 hit) with verification artifact `d9e73413`; the verifier script itself has SHA-256 prefix `017d1e44`. Proposition 7.19 is a negative structural datum: it shows that no purely local argument — circuit axioms, covered rotating owner, triangle-freeness of the atom graph — can close twenty-five atoms, since all of these conditions are simultaneously realizable. Both circuits are nevertheless excluded, by the two certificate layers that follow. Throughout, *the* circuit #298 (respectively #264) refers to the exhibited witness of Proposition 7.19: the atom set and selection of the pinned rebuild shipped in the ancillary files (`anc/t5_artifacts/`), over which the mechanical certificates and the ambient exclusions below quantify. The by-hand proof of Proposition 7.20 is independent of this choice: it applies verbatim to every twenty-five-atom circuit on the printed graphs at the recorded pairs.

Proposition 7.20 (Dead owners). *In both circuits, every selection realizing the covered rotating owner at the recorded pair (v, x_0) leaves the owner dead.*

Proof. Circuit #298, $(v, x_0) = (0, 17)$. From the printed string, $\deg_F(17) = 2$ with $N_F(17) = \{0, 1\}$. Let ω realize the owner. Each covering row for a star pair $\{17, y\}$ contains 17 and avoids 0 (Lemma 7.7), so its edges at 17 lie in $\{\{1, 17\}\}$; hence the edge $\{1, 17\}$ is selected. In the latent graph minus v , the vertex 17 is therefore isolated: the tail is $\{17\}$. Since 17 lies in the class opposite v , it is not an atom-partner of v , and no atom lies inside a single vertex. The owner is dead. Moreover $N_F(0) = N_F(1) = \{9, 10, 11, 13, 17\}$, so the transposition of 0 and 1 is an automorphism of F and the same argument disposes of the mirror profile at $(1, 17)$.

Circuit #264, $(v, x_0) = (0, 9)$. From the printed string, $\deg_F(9) = 3$ with $N_F(9) = \{0, 1, 2\}$, and $N_F(0) = \{9, 10, 12, 13, 15\}$, so the star pairs are $\{9, y\}$ with $y \in \{10, 12, 13, 15\}$. Exhaustive inspection of the full geodesic database of F (a finite check on the printed graph) shows:

- (f1) every row of *any* same-class distance-four pair that contains 9 and one of 10, 12, 13, 15 and avoids 0 uses the edge $\{1, 9\}$ (the pairs $\{9, 10\}$, $\{9, 12\}$, $\{9, 13\}$ each have exactly three such rows, $\{9, 15\}$ exactly one);
- (f2) the unique such row through 9 and 15 is $(15, 2, 9, 1, 17)$, a row of the pair $\{15, 17\}$, and it uses the edge $\{2, 9\}$ as well.

Let ω realize the owner. Covering rows avoid 0 (Lemma 7.7), so by (f1) the edge $\{1, 9\}$ is selected, and by (f2) the covering row for $\{9, 15\}$ must be $(15, 2, 9, 1, 17)$ — in particular $\{15, 17\} \in \mathcal{A}$ — so $\{2, 9\}$ is selected too. Thus both edges at 9 other than the active edge $\{0, 9\}$ are selected, the tail is $\{9\}$, and the owner is dead as before. $\square$

The database facts (f1), (f2) are also certified mechanically: per-edge unsatisfiability of the alternative (`verify_t5_tail_blanket_unsat.py`, artifact 3720a8c2). Independently of Proposition 7.20, the full statement “over all admissible row selections, no atom has both endpoints in the latent component of the owner” was checked as a monolithic constraint system twice per circuit, by CP-SAT infeasibility and by a CaDiCaL replay of a CNF encoding (for #298: 1680 variables, 5239 clauses); artifact prefixes `f5c0cbca` and `a8a160d5` (#298), `79471ef0` and `c7fbcc70` (#264). These certificates are *intrinsic*: they quantify over selections of rows of the circuit itself. Graph-level realizations, whose ambient crossing graph could a priori supply further structure, are addressed by the second layer.

Proposition 7.21 (Computer-assisted; ambient exclusion of #264). *There is no triangle-free graph H on at most 25 vertices with a two-colouring of $V(H)$ extending the classes of circuit #264 such that: the crossing edges of H contain $E(F)$; the monochromatic edges of H are exactly the 25 atoms; for every atom $\{x, y\}$ the distance of x, y in the crossing graph of H is four and every length-four crossing geodesic is a row of the circuit; and the two-colouring is a maximum cut of H . The eight constraint systems (one for each assignment of $k \in \{0, \dots, 7\}$ added vertices to the owner class, unused vertices being permitted to be isolated) are all infeasible.*

The primary run (`extend_t5_hit_maxcut.py`) separates the maximum-cut switch inequalities exactly and lazily; an independent SAT replay (`verify_t5_maxcut_extension_unsat.py`) reproduces all eight infeasibilities from the unrestricted crossing-edge domain. The obstruction is finite and explicit. The vertex set $S = \{4, 5, 6, 7, 8, 11, 14, 16\}$ of the circuit is crossed by 23 atoms and only 2 support edges, so switching S forces any row-preserving maximum-cut extension to add at least 21 crossing edges on the boundary of S ; exact counting of addable, row-preserving edges gives capacities 21, 19, 17, 15, 13, 11, 9, 7 across the eight assignments, killing seven of them outright, and the single assignment with capacity 21 is killed jointly by a second switch (joint demand 42 against joint capacity 28).

Proposition 7.22 (Computer-assisted; ambient exclusion of #298). *There is no triangle-free graph H on at most 25 vertices with a two-colouring of $V(H)$ extending the classes of circuit #298 such that: the crossing edges of H contain $E(F)$; the monochromatic edges of H are exactly the 25 atoms; for every atom $\{x, y\}$ the distance of x, y in the crossing graph of H is four and every length-four crossing geodesic is a row of the circuit; and the two-colouring is a maximum cut of H . The eight constraint systems (one for each assignment of $k \in \{0, \dots, 7\}$ added vertices to the owner class, unused vertices being permitted to be isolated) are all infeasible.*

The same two verifiers as for #264 agree on all eight infeasibilities. Here the displayed two-colouring already fails to be a maximum cut of the bare graph $(V(F), E(F) \cup \mathcal{A})$: the vertex set $S = \{4, 5, 6, 7, 8, 12, 14, 15, 16\}$ of the circuit is crossed by 24 of the 25 atoms and only 4 support edges, so switching S forces any row-preserving maximum-cut extension to add at least 20 crossing edges on the boundary of S . Unlike for #264, this demand alone kills no assignment; the infeasibilities are joint with the row-preservation clauses and, at $k \in \{0, 1\}$, a second switch. The certificate chain for #298 is included in the ancillary files (`anc/t5_artifacts/298.extension/`).

7.6 Status of orders seventeen through twenty-one

Proposition 7.23 (Computer-assisted). *At order 17, the systems $(R_{7,10})$, $(R_{8,9})$ and $(R_{12,5})$ are infeasible (artifact prefixes `4d450ca0`, `f76b9d64`, `325012ec`, in the split order listed, the mapping being fixed by the bit-exact regeneration in the ancillary files; single-solver certificates, as in Proposition 7.18). The systems $(R_{9,8})$, $(R_{10,7})$, $(R_{11,6})$ are feasible.*

Feasibility of a support relaxation is not existence of a rotor: at each of the three feasible splits a staged search of 3,000 support graphs produced no circuit realizing a live covered rotating owner (artifacts 95dbc901, d612c59e, 31d82c62), and at order 18 the first 350 supports at split (9, 9) were likewise all rejected at the live-owner gate. These are bounded searches and prove nothing beyond the instances examined. Orders 19–21 have not been examined. The honest summary is:

- orders ≤ 14 : impossible for any covered rotating owner (Theorem 7.12), by hand;
- orders 15, 16: impossible for balanced rotors (Proposition 7.18), computer-assisted;
- order 17: open; owner-class size restricted to $\{9, 10, 11\}$ (Proposition 7.23);
- order 18: open; the only circuits found so far that pass all local tests are #298 and #264, and both are excluded by certificates (Propositions 7.20, 7.21 and 7.22);
- orders 19–21: open and unexamined; order 22 and beyond: impossible (Corollary 7.10).

All evidence is consistent with the following statement, which we can neither prove nor refute.

Definition 7.24. Say that a support circuit at $t = 5$ is *scope-vacuous* if every selection realizing a covered rotating owner, at every pair (v, x_0) , leaves the owner dead.

Open question. *Is every support circuit at twenty-five atoms scope-vacuous? In particular, is there no balanced rotor at $t = 5$?*

A positive answer would close the parameter $t = 5$ of the programme in the same sense as the finite closures at $t = 3$ and $t = 4$; by Remark 7.15 it would still be one more finite base case, not a statement for all t . A negative answer — a live covered rotating owner on a triangle-free 25-atom circuit of order 17–21, surviving an ambient extension test — would refute the rotor route in its present form. Neither outcome, we stress, resolves the Erdős inequality $\text{bip}(G) \leq N^2/25$: the results of this section constrain one local proof strategy and produce no new bound on $\text{bip}(G)$.

8 Reproducibility

The ancillary archive (`anc/`) contains every verifier script named in this paper, the production sources of every cited Lean 4 module, a command-level `README`, and a `SHA256SUMS` manifest; the hashes below are SHA-256 and match that manifest. The Lean modules belong to the development `Erdos23Delta0`; the declarations cited in Sections 3, 5 and 6 compile without `sorry` or `native_decide`, with axioms within `{propext, Classical.choice, Quot.sound}`; for `LiveMiddleSwapCrossOuter.lean` this axiom audit is attested twice: by the acceptance ledger of the source archive and by a fresh rebuild of its five-module import chain with a kernel `#print axioms` probe, which reports axioms exactly `{propext, Quot.sound}` (`anc/lean_axiom_probe/`, verdict `PASS_AXIOM_PROBE`; Proposition 6.14).

fact	verifier(s)	artifact SHA-256
order 15, 4 splits	<code>rooted_t5_support_cp_sat.py</code>	14ac9d93 8d1946
order 16, 5 splits	<code>rooted_t5_support_cp_sat.py</code>	8dff49ea bd0c03
order 17, 3 splits	<code>rooted_t5_support_cp_sat.py</code>	4d450ca0 f76b9d
#298 circuit + owner	primary + <code>verify_t5_local_classifier_hit.py</code>	c1d474d7, verifica
#264 circuit + owner	primary + <code>verify_t5_local_classifier_hit.py</code>	6595501f, verifica
#298 all-row dead owner	CP-SAT + <code>verify_t5_active_scope_unsat.py</code>	f5c0cbca, a8a160
#264 all-row dead owner	CP-SAT + CaDiCaL replay	79471ef0, c7fbc0
#264 facts (f1), (f2)	<code>verify_t5_tail_blanket_unsat.py</code>	3720a8c2
#264 ambient exclusion	<code>extend_t5_hit_maxcut.py</code> + <code>verify_t5_maxcut_extension_unsat.py</code>	eight infeasibilitie
#298 ambient exclusion	<code>extend_t5_hit_maxcut.py</code> + <code>verify_t5_maxcut_extension_unsat.py</code>	source ff1c1209,

Table 3: Computer-assisted certificates of this section. Primary searches use OR-Tools CP-SAT; independent replays use exact graph operations (NetworkX) or CaDiCaL via PySAT. All arithmetic is integer or Boolean. The printed prefixes are prefixes of each artifact’s embedded canonical payload hash. The historical artifacts were session records of the programme ledger; on 2026-07-17 the set was regenerated from the archived scripts and is shipped in the ancillary files (`anc/t5_artifacts/`, with a per-artifact record). The twelve split certificates regenerate bit-exactly to the twelve listed hashes. The circuit-level artifacts depend on a solver-nondeterministic enumeration order and do not replay byte-exactly; both circuits were instead rebuilt on the printed support graphs at the recorded (v, x_0) pairs (`rebuild_t5_local_classifier_hit.py`) and every downstream verdict was reproduced — in particular the #298 scope certificate reproduces at exactly 1680 variables and 5239 clauses. The historical prefixes are retained above as the session record. In an independent adversarial review pass (2026-07-17), the twelve infeasibility and three feasibility verdicts of Propositions 7.18 and 7.23 were re-run from the archived model, the two hits of Proposition 7.19 were re-derived with an independently written constraint encoding, and the finite facts of Proposition 7.20 were re-verified by exhaustive geodesic enumeration.

Lean 4 sources (`anc/lean/`)

```

SingletonPairSigma.lean
  e4060bcc9b8b92dc0f65d7c4022bf2d02a7e172c9210e3e5ee10b6e273383793
CutTightStarPigeonhole.lean
  dd6da23c53e426f1ef10a7a1165508b2d519d562f99227b24bf19c823d860758
R43SupportIncidence.lean
  6664967000f45660bd82c9ea9304942257248fbca5bb88482c45f6397d184cea
BadStarCoverFreeness.lean
  afd944ea407a6fb29a9e005c4827b7e8af9e12edc7dcc532a9bd523e083ae253
K33BadStarPairCountZero.lean
  354565234093e4e25b7a4f296b985a43a1809b438f0633c7c97f7fe422e33233
CutTightActiveRotorIncidence.lean
  0814a665869737b86a93b44b7e42c55690b81058e221891c36b26fc168b688f1
SaturatedRotorSupportPersistence.lean
  b4438520f7a7ac7dee4d8ff3587cd85bfa3312ee286649001d3fe6cbea82e18d
R44K2TailOverlap.lean
  12dfb92724f36beac1f11eba529023f0445893ca6e60376b09b2403c0e64066a
LiveMiddleSwapCrossOuter.lean
  3dff7897f65112f4f8177b84ab28f631c85bc6f50f7f5d920ed06f033b7f9275

```

Verifier scripts (`anc/`)

Eight-vertex rotor (Section 4); exhaustive, rerun for this paper with verdict CLAUDE-GATE=PASS (exhaustive):

```
_claude_r39_8vtx_rotor_gate.py
6d74bcbdb1bab12948c5e1a498f62a7185b03743a2b701ec5aeba6f54b01b2aeb
```

Sixteen-atom census (Section 6); primary pipeline, second acceptance path, and the abstract-circuit pair:

```
enumerate_t4_support_graphs.py
540cbdb90faba872a359285be9cea7b622410cffdde44316443cf7b4f4fd96d1
enumerate_t4_atom_circuits.py
7ec51265a3b4b3e8c9fa016561259795086bc05464e692d4179f28afe57702b5
enumerate_t4_profile_transitions.py
c88318342312b12c70c52e638650450d4b21352c1724a9bf9ab97ec08c88d749
verify_t4_support_census.py
430b71893833040f2da8de1e1aa447d280e8845218e4ed411e0e0a4c635de88a
verify_t4_atom_census.py
535861568f27112e437546e64242f6ad03f410342b15cb5ff5f1d3b18f979e6a
verify_t4_profile_exclusion.py
0a3c0c02b240183ea87b2acf733b62e8966d7907f845fa289daacc381c587450
verify_t4_cross_outer_exclusion.py
657873636f1e509b9c2c7e0da2b36a2a3231dfd85d330b9d2ff28254fae1ea31
search_t4_support_circuit.py
e0f19a43e6f7e922fc55e743fdb3bc4a1e02a37a5cc90fafea38aff1b0cb35cd
verify_t4_support_circuit.py
4b4458a497001c1696ee512e0cb1ec4e51b859e91bd7bfab441e56040f821f42
```

The five canonical JSON artifacts of the census are included in `anc/t4_artifacts/` and carry the embedded canonical hashes printed in the verification record of Section 6; the entire pipeline, all five artifacts, and the `geng` counts of Table 2 were regenerated bit-exactly from the archived scripts on 2026-07-17.

Twenty-five-atom certificates (Section 7):

```
rooted_t5_support_cp_sat.py
dfd060bd1955e7fda6982958f40fdb70a1504a49827a7dd1efaa77e89df03053
verify_t5_local_classifier_hit.py
017d1e44e6dbacab39b82d99c451e9154afa7dfc560a0c478a28d8b778153aa6
verify_t5_active_scope_unsat.py
7a042c06edf61fe35a561e1302e6656e0a09118bfba2052f811fd0e1eb7ca154
verify_t5_tail_blanket_unsat.py
566f218709b3c5d32cb0e15a4ff8e985ce30479535ca264c319456ac2d59b311
extend_t5_hit_maxcut.py
b0e362ac457171eb16e01260cd9af868cdc25e2fa20cac6445730c74c09934a2
verify_t5_maxcut_extension_unsat.py
ebbf48bfc7b7ad14fc5a594b92c9be5132cb7fbb015c0c1ad1880ba380c1128e
rebuild_t5_local_classifier_hit.py
34859ea68026b2f860115af71e66884685d834cbf10eb7d321f1ed4ee2a151a1
check_t5_active_scope_profile.py
11a4126ddfca0cc786423cf7b618b42ab2f983970e46f1cb5ac3abf7cd677221
```

The $t = 5$ certificate *artifacts* cited by SHA-256 prefix in Table 3 were regenerated from the archived scripts on 2026-07-17 and are shipped under `anc/t5_artifacts/` together with a regeneration record (`NOTES_t5_regeneration.md`): the twelve split-infeasibility certificates regenerate bit-exactly to the cited hashes, and the circuit-level certificates were rebuilt on the two printed support graphs with identical verdicts, the historical prefixes remaining as the session record. In an independent adversarial review pass on 2026-07-17, the twelve infeasibility and three feasibility split verdicts were re-run from the archived model, both circuit hits were re-derived with an independently written constraint encoding, and the finite facts behind Proposition 7.20 were re-verified by exhaustive geodesic enumeration. The nine order-15/16 certificates and the three order-17 certificates are single-solver (OR-Tools CP-SAT) results, as flagged in the text. Graph generation throughout uses `geng` from `nauty/Traces` [9]; no floating-point arithmetic is used on any acceptance path.

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